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खान मंत्रालय**

**GOVERNMENT OF INDIA
MINISTRY OF MINES**

**भारतीय भूवैज्ञानिक सर्वेक्षण
GEOLOGICAL SURVEY OF INDIA**

**महाराष्ट्र के जिला चन्द्रपुर के गोविंदपुर ताम्र खनिज प्राप्ति के आसपास ताम्र तथा
उससे जुड़े स्वर्ण खनिजन के क्षेत्रों की स्थापना हेतु अन्वेषण पर प्रतिवेदन
(पी-II स्तर)**

(मद कोड सं. ०३९/एमआईपी/सीआर/एमएच/२००७/०१२)

(कार्यसत्र २००७ - २००९)

**REPORT ON INVESTIGATION FOR ESTABLISHING ZONES OF COPPER
AND ASSOCIATED GOLD MINERALISATION AROUND GOVINDPUR
COPPER PROSPECTS, CHANDRAPUR DISTRICT, MAHARASHTRA
(P-II STAGE)**

(Item code no. 039/MIP/CR/MH/2007/012)

(Field Season 2007 - 2009)

द्वारा

By

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**नागपुर २०१०
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(PART OF TOPOSHEET NO. 55P/11)
Field Seasons 2007-08 and 2008-2009
(Item Code No. 039/MIP/CR/MH/2007/012)

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महाराष्ट्र के जिला चन्द्रपुर के गोविंदपुर ताम्र खनिज प्राप्ति के आसपास ताम्र तथा उससे जुड़े स्वर्ण खनिजन के क्षेत्रों की स्थापना हेतु अन्वेषण पर प्रतिवेदन

(पी-II स्तर)

(धरातल पत्रक सं ५५पी/११ के भाग)

क्षेत्रकाल २००७-२००८ तथा २००८-२००९

(मद कोड सं. ०.३९/एमआईपी/सीआर/महा./२००७/०१२)

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सारांश

ताम्र खनिजन के निरूपण क्षेत्र के उद्देश्य से वर्ष २००७-२००९ के दौरान १:२,००० पैमाने पर ०.५ वर्ग किलोमीटर क्षेत्र का विस्तृत मानचित्रण गोविंदपुर तथा बोथली क्षेत्र के आसपास किया गया, जिसमें क्रमबद्ध नमूने ५१६ मृदा प्रतिदर्शी, ५५ संस्तर शैल प्रतिदर्शी, ११ गर्त प्रतिदर्शी तथा १० शैलकीय प्रतिदर्शी पर कार्य किया गया। इस गवेषण क्षेत्र में पेनगंगा समूह के अवसादी आवरण शैलों के साथ आर्कियन आधारी शैलें शामिल हैं। यह आधारी शैलें मेटापेलाइटों के साथ बेंगपाल नाइस द्वारा प्रतिनिधित्व करते हैं जो कि पेग्मेटाइट, क्वार्टज तथा एपिडोट शिराओं द्वारा अन्तर्वेधित हैं।

गोविंदपुर पुराने कार्य क्षेत्र में मेटापेलाइटों, क्वार्टज, अभ्रक शिस्ट के अन्तर्वेधन से मेलाकाइट धब्बों के साथ क्वार्टज शिराओं को मानचित्रित किया था। पुराने कार्य क्षेत्र में और उसके आसपास क्वार्टज टूरमैलीन पट्टियों के कुछेक छोटे वियुक्त दृश्यांश भी देखे गए हैं। उत्तर-दक्षिण प्रवणता की खनिजीकृत क्वार्टज शिरा उत्तरपूर्व की ओर अल्प अवनमन निम्न निमज्जन (१२.१५) के साथ स्पष्ट रूप से वलनित है।

विकीर्णन के रूप में, बोथली सल्फाइड (चाल्कोपायराइट, बोर्नाइट तथा पायराइट) सिलिसीकृत क्षेत्रों में देखा गया है।

सिलिसीकृत क्षेत्र में मेलाकाइट पर्पटी द्वारा ताम्र खनिजन दिखाई देती है जो कि बेंगपाल नाइस के भीतर मेटापेलाइट के परिक्षेत्र को परिसीमित करती है तथा इनकी प्रवणता उत्तर-दक्षिण तथा उत्तर-उत्तरपूर्व - दक्षिण-दक्षिणपश्चिम की ओर भ्रंश क्षेत्रों की ओर अभिस्थापित है। क्षेत्रकाल वर्ष २००८-२००९ में ताम्र खनिजन के विक्षेपण पद्धति को जानने

के लिए १०० मीटर x २५ मीटर गिड पद्धति में कुल २७६ मृदा प्रतिदर्शों को एकत्रित किया गया।

गिड पद्धति (२५ x १०० मीटर) में गोविंदपुर तथा बोथली क्षेत्रों में मृदा एवं संस्तर शैल प्रतिदर्शों दोनों को एकत्रित किया गया। प्रतिदर्श १०० क्षिद्र आकार में शुष्क तथा जालीदार है। मृदा प्रतिदर्शों का विश्लेषण मान २० पीपीएम से ०.६४% तक ताम्र, <१० से ८०० पीपीएम तक सीसा, २० से ६४० पीपीएम तक जस्ता, तथा संस्तर शैल प्रतिदर्शों के लिए मान <१० पीपीएम से ३.१५% ताम्र, <१० पीपीएम से ६० पीपीएम सीसा, <१० पीपीएम से ०.१८% जस्ता है।

गोविंदपुर पुराने कार्यक्षेत्र के निकट से एकत्रित किए गए (६) मृदा नमूने विसंगति क्षेत्र में ०.१०% से ०.६४% तक ताम्र श्रेणी सूचित करता है तथा उसकी नतिलंब लंबाई १२५ मीटर है।

अध्ययन क्षेत्र से एकत्रित किए गए सरिता अवसाद नमूनों में १२ पीपीएम से ११६ पीपीएम तक ताम्र, ९ पीपीएम से ७५ पीपीएम तक सीसा, १७ पीपीएम से ६२ पीपीएम तक जस्ता, ७ पीपीएम से १४६ पीपीएम तक निकेल तथा स्वर्ण मात्रा <१०० पीपीबी मिलती है।

गोविंदपुर क्षेत्र में गर्त/खाई के विश्लेषणात्मक मान ताम्र श्रेणी में <१०० पीपीएम से ८९५ पीपीएम तक, सीसा ७ पीपीएम से ४१ पीपीएम तक, जस्ता ३२ पीपीएम से १६९ पीपीएम तक, कोबाल्ट २० पीपीएम से ७५ पीपीएम तक, निकेल ३० पीपीएम से १२३ पीपीएम तक है। ४ गर्त/खाई नमूनों में स्वर्ण का मान १०० पीपीबी से १४० पीपीबी तक मिलता है जबकि अन्य गर्त/खाई में स्वर्ण मान <१०० पीपीबी मिलता है।

बोथली क्षेत्र में गर्त/खाई के नमूनों का विश्लेषणात्मक मात्रा की श्रेणी १० पीपीएम से २०० पीपीएम तक ताम्र, १० पीपीएम से २० पीपीएम तक सीसा, १०० पीपीएम से १२५ पीपीएम जस्ता, ४० पीपीएम से ७५ पीपीएम तक कोबाल्ट, <१० पीपीएम से १० पीपीएम तक निकेल, २ पीपीएम से १० पीपीएम तक आर्सेनिक है जबकि स्वर्ण <१०० पीपीएम तक मिलता है तथा संस्तर शैल नमूनों में <१० पीपीएम से ६० पीपीएम तक सीसा, <१० पीपीएम से १७५ पीपीएम तक जस्ता, ४९ पीपीएम से २७५ पीपीएम तक कोबाल्ट, <१० पीपीएम से २० पीपीएम तक आर्सेनिक तथा <१०० पीपीबी से १६० पीपीबी तक स्वर्ण मिलता है।

शीतल निष्कर्षण पद्धति को मृदा नमूनों के लिये अपनाने के बाद ताम्र मूल्य के निष्कर्षण के लिए उपयोग में लाया गया। कुल ७८ मृदा नमूनों का विश्लेषण

किया गया तथा २५ मृदा नमूनों में (३००० पीपीएम से ६००० पीपीएम तक) आधार धातु का पता लगाया गया ।

वर्तमान क्षेत्रकाल के दौरान खनिजन के मूल को समझने के लिए विस्तृत पेट्रोमाइक्रोस्कोपी अध्ययन किया गया । अध्ययन क्षेत्र में, यंगर खनिजन अपरूपित क्वार्ट्ज शिराएं / रीफ के साथ एम्फिबोलाइट, फाइब्रोलाइट, माइलोनाइट तथा क्वार्ट्ज-क्लोराइट माइलोनाइट ये उपस्थित प्रमुख शैल प्रकार हैं । इस क्षेत्र में माइलोनाइट तथा क्वार्ट्ज रीफ में तीव्र माइलोनीकरण तथा विखंडन से यह पता चलता है कि यहां भंगुस्तन्य अपरूपण हुआ है । जलोढक अपरूपित क्वार्ट्ज शिराएं/ रीफ द्वारा सल्फाइड खनिजन जैसे पायराइट, चाल्को पायराइट, कोवेलाइट तथा पेक्साइट आतिथेय / परपोषी हैं ।

इस क्षेत्र में खनिजन की तीन भिन्न भिन्न प्रावस्था पाई जाती हैं । खनिजन की इन सभी प्रावस्थाओं को अपरूपण की विभिन्न अवस्थाओं तथा क्वार्ट्ज रीफ / शिरा अन्तर्वेधन के साथ सह-सम्बद्ध किया जा सकता है । सूक्ष्मदर्शी द्वारा क्वार्ट्ज में संस्तरों का अवलोकन करने से यह पता चलता है कि क्वार्ट्ज बहुमूल्य सिलिका-समृद्ध जलोश्म द्रव्य से अवक्षेपित है । सिलिका-समृद्ध जलो में सल्फाइड खनिजन पाया जाता है ।

क्षेत्रकाल २००७-०८ में महाराष्ट्र के जिला चन्द्रपुर के गोविंदपुर खण्ड में ०.८५ वर्ग किलोमीटर के क्षेत्रफल में भूभौतिकी सर्वेक्षण में नियोजित एसपी, चुम्बकीय, प्रतिरोधक के साथ आईपी तकनीक (ग्रेडिएन्ट ऐरे), वर्टिकल इलेक्ट्रिकल साउंडिंग तथा डाइपोल - डाइपोल द्वारा किया गया ।

**REPORT ON INVESTIGATION FOR ESTABLISHING ZONES OF
COPPER AND ASSOCIATED GOLD MINERALISATION AROUND
GOVINDPUR COPPER PROSPECT, CHANDRAPUR DISTRICT,
MAHARASHTRA**

(P-II STAGE)

(PART OF TOPOSHEET NO. 55P/11)
Field Seasons 2007-08 and 2008-2009
(Item Code No. 039/MIP/CR/MH/2007/012)

By

R.H. Chavhan, Jr. Geologist, K. C. Mahapatra, Sr. Geologist (PT),
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B.V. Srirama, Jr. Geophysicist

PART-I

ABSTRACT

Detailed mapping of 0.5 sq km area on 1:2,000 scale, systematic sampling, 535 soil samples, 90 bed rock samples, 5 nos. of pit samples and 25 petrological samples, in grid pattern around Govindpur and Bothli areas were carried out during 2007-09 with the objective to delineate zones of Copper mineralisation. The explored area comprises Archaean basement rocks with sedimentary cover rocks of Penganga Group. The basement rocks are represented by Bengpal gneiss with metapelites, which have been intruded by pegmatite, quartz and epidote veins.

Quartz vein with malachite stains intruding metapelites (quartz mica schist) was mapped in Govindpur old working area. A few small isolated outcrop of quartz - tourmaline bands are also seen in and around old working area. The mineralised quartz vein trending N-S is broadly folded with low plunge (12-15°) towards NE.

At Bothli Sulphides (chalcopyrite, Bornite and pyrite), in the form of disseminations, are noticed in the silicified zones.

The copper mineralisation is manifested by malachite encrustations in silicified zone, which are confined to enclaves of metapelite within Bengpal gneiss and have been emplaced along N-S and NNE-SSW trending fault zones. A total of 325 soil samples were collected in grid pattern 100m x 25m in order to know the dispersion pattern of copper mineralisation during field season 2008-09.

Both soil and bed rock samples were collected in Govindpur and Bothli areas in grid pattern (100 x 25). The analytical values in soil samples range from 20 ppm to 0.10% Cu, <10 to 125 ppm Pb, 20 to 500 ppm Zn whereas the values range from 0.77% to 3.15% Cu, <10 to 0.18% Pb and <10 to 100 ppm Zn for bed rock samples.

Soil samples (6 nos) collected near Govindpur old working, show anomalous zone indicates Cu values range from 0.10% to 0.64% of strike length 125m.

Stream sediments samples collected from the study area show Cu values range from 12 ppm to 116 ppm, Pb from 9 ppm to 75 ppm, Zn from 17 ppm to 62 ppm, Co from 7 ppm to 178 ppm, Ni from 17 ppm to 146 ppm, and Au values <100 ppb.

In Govindpur area, the analytical values of the pits/trenches samples shows copper values range from <100 ppm to 895 ppm, Pb values from 7 ppm to 41 ppm, Zn from 32 ppm to 169, Co from 20 ppm to 75 ppm, Ni from 30 ppm to 123 ppm, Au values of 4 pits/trenches ranging from 100 ppb to 140 ppb while others 2 pits/trenches shows Au values <100 ppb.

In Bothli area, the analytical values of the pits/trenches samples range from 10 ppm to 225 ppm Cu, 10 ppm to 20 ppm Pb, 100 ppm to 125 ppm Zn, 40 ppm to 75 ppm Co, <10 ppm to 10 ppm Ni, 2 ppm to 10 ppm As while Au is <100 ppb and for bed rock samples from <10 ppm to 0.25% Cu, <10 ppm to 60 ppm Pb, <10 ppm to 175 ppm Zn, 40 ppm to 275 ppm Co, <10 ppm to 20 ppm Ni, <1 ppm to 10 ppm As, and Au values <100 ppb to 160 ppb.

The cold extraction method have been adopted and used for determination of Cu% and other elements in ppm in the soil as well in the stream sediments samples. The total 78 soil samples were analysed and detected base metal in 25 soil samples (ranges from 3,000 ppm to 6,000 ppm).

During the current field season detailed petromicroscopic studies has been carried out to understand the origin of mineralization. The main rock types present in study area are quartz-chlorite mylonite, quartz-fibrolite mylonite, garnetiferous quartz monzonite, and amphibolites with younger mineralized sheared quartz veins/reef. Intense mylonitisation and brecciation in mylonite and quartz vein/reef indicates brittle-ductile shearing occurred in the area. The sulphide mineralization viz., pyrite, chalcopyrite, covellite and paxite are hosted by hydrothermal sheared quartz veins/reef. Three distinct phases of mineralization are present in the area. All these phases of mineralization can be correlated with different stages of shearing and quartz reef/vein intrusion. The zoning in quartz observed under microscope suggests that quartz precipitated from a polygenic silica-rich hydrothermal fluid. Silica-rich hydrothermal fluid carries the sulphide mineralization.

Geophysical surveys employing SP, Magnetic, resistivity cum IP techniques (Gradient Array), Vertical electrical sounding and Dipole-dipole were conducted over an area of 0.85 Sq km in Govindpur block of Chandrapur District of Maharashtra in FS 2007-08

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1.0 INTRODUCTION

1.1 General:

In pursuance of FSP item code No. 039/MIP/CR/MH/2007/012, page no. 48 and 49 of Geological Survey of India, Central Region, investigation for copper, involving detailed Mapping 0.5 sq km (1:2,000 Scale), large scale mapping (150 sq km), soil and stream sediment sampling (516 nos.), pitting / trenching, collection of bed rocks samples (55 nos.), was carried out in southeast and western part of Govindpur Village, Chandrapur District, Maharashtra, falling in Survey of India Toposheet No. 55P/11. The area is bounded by Latitudes $20^{\circ}28'23''$ - $20^{\circ}28'45''$ and Longitude $79^{\circ}36'55''$ - $79^{\circ}37'38''$ for Govindpur block and Latitudes $20^{\circ}28'08''$ - $20^{\circ}28'37''$ and Longitude $79^{\circ}32'03''$ - $79^{\circ}32'52''$ for Bothli block. Large scale mapping was carried out on 12,500 scale and is bounded by Latitudes $20^{\circ}23'27''$ - $20^{\circ}30'$ and Longitude $79^{\circ}30'$ - $79^{\circ}38'02''$.

The layout of geophysical traverses at Govindpur copper prospect, falling in the Survey of India toposheet No. 55P/11, along with geology (Plate II). The N08°E- S08°W oriented 0.85 km baseline was laid during the field season 2007-08. The N82°W- S82°E oriented seventeen traverses between N60 and S20 were laid at 100m separation orthogonal to baseline drawn with respect to a reference point (0/0). These traverses laid over a length of 0.8 km were used to carry out geophysical surveys at a station interval of 10m between stations W20 and E30 on either side of the baseline. An area of 0.85 sq km was covered by geophysical surveys during the field season 2007-08.

1.2 Location and Accessibility:

The area in Talodi tehsil of Chandrapur district is about 120 km from Nagpur. It is approachable by State Highway No. 9B to Talodi from where 9 km metalled road join to Govindpur Prospect area. The road leads onward to Chimur via Neri. Bothli is accessible by jeepable road and is at a distance of 10 km from Govindpur area. The nearest railway station is Nagbhir. Nagpur –Nagbhir-

Chandrapur extension line of Southeastern railway (0.76m narrow gauge) passes through Talodi road and Sindewahi road railway stations.

1.3 Physiography and Drainage:

Physiographically, the area exhibits a pediment zone/peneplain in major part while minor structural hills and valleys are also seen. Southerly flowing Upasa nala and Gondri nadi form the tributaries of Wainganga River. Ghodazhari canal system irrigates the paddy field in this area. The area around Bothli in general is a flat terrain except for two hills comprising younger sedimentary rocks (equivalent to Vindhya) in the eastern and other in western parts of the area. The area is drained by Gondri Nadi, east of Bothli and also by number of southeasterly nalas. Water level in the wells range from 3 to 9m in depth from surface.

1.4 Climate and vegetation:

The climatic conditions in the areas are extreme both in summer and winter. In summer temperature ranges from 38⁰ to 45⁰C while during winter it drops to a minimum of 12⁰C. The average rainfall in the area is about 125 cm, much of which precipitated between July and August.

The area is thickly forested by open and bushy, mixed open forest comprises of Bhramapuri and Pendhri reserved forest to the west and north of Govindpur respectively.

Teak, Bija, Tendu, Bamboo etc. are the most common trees noticed in the area.

1.5 Previous work:

The area covering 55P/11 was mapped on 1:63,360 scale by Parimoo, M.C. and Nag, P (1959-60). Gneisses with migmatites, metasedimentary enclaves and younger sedimentary cover, equated to Vindhya were mapped in the area.

During the FS 1971-1974 Directorate of Geology and mining carried out prospecting for copper mineralisation about 1.8 km east of village Bothli. Directorate of Geology and mining open a large pit of 30mts length, width 25m and maximum depth of 6m near its western wall, filled with quartz and silicified sheared quartz mica schist enriched with malachite.

Two underground openings are located in this silicified zone at the bottom of the western wall, are vertical and 2m in diameter. This silicified zone comprises of copper, cuprite and malachite.

Two small circular shafts each about 4 mts in diameter and 6m depth was located to the east of the northeastern corner of the large pits in Govindpur old working.

1.6 Present work

Detailed geological mapping was carried out covering an area of 0.5 sq km on 1:2,000 scale in Toposheet No. 55P/11 around Govindpur and Bothli.

A map of Govindpur and Bothli areas was prepared on 100 x 25m grid, where geophysical chargeability zone has been identified and plotted all the geological features such as strike, dip, lithological boundaries etc.

535 nos. soil samples have been collected in Govindpur and Bothli areas on a grid pattern of 100 x 25m to know the dispersion pattern of copper mineralization and 90 nos bed rock samples were collected in both areas indicate encrusted malachite with chalcopyrite and cuprite.

A total of 200 cum pitting/trenching were carried out in both the areas to delineate extension of copper ore body. A total 50 nos. of pitting/trenching samples were also collected during this field season. 25 nos. of representative samples from Govindpur and Bothli area were collected for petrological studies to understand control of mineralization

During field season 2007-2008, besides sampling, large scale mapping on 1:12,500 scale was carried out covering 150 sq km Six pits/trenches in Govindpur block and four pits/trenches in Bothli areas were excavated. 211 soil samples, 29 nos. stream sediments samples, were collected and analyzed. Rapid analyses of 78 nos. soil samples by cold extraction method were carried out during FS 2007-08.

Nature and quantum of work and time schedule:

Nature of work	Total workload envisaged	Work completed during 2007-08	Work completed during 2008-09	Progress Target Achieved	Remarks
1. LSM (1:12,500)	150 sq km	150sq km	Nil	150sq km	
2. DM (1:2000)	0.5 sq km	Nil	0.5 sq km	0.5 sq km	
3. SMPL a. SSS b. SS c. BRS d. P/T e. PTS f. PS 4. Chem. Analy CU, Pb, Zn, Ni, Co, Au, Ag, As. 5. GP (IP, SP, Res.)	100 nos 535NOs. 90nos. 200 Cu m 50Nos. 25nos. 775 nos. 9 Lkm.	29nos. 211Nos. 06 nos. 100Cu m 7 nos. Nil 253 nos. 9 Lkm	Nil 276 Nos. 49Nos. 100Cu m 4Nos. 10 nos. 329 nos. Nil	29 Nos. 487 Nos. 55 Nos. 200 Cu m 11 Nos. 10 nos. 582 nos. 9 Lkm.	78nos soil samples rapid analysis by cold extraction method.

Note: Large Scale Mapping (LSM), Detailed Mapping (DM), Stream Sediment Samples (SSS), Soil Samples (SS), Bed Rock Samples (BRS), Pitting/Trenching (P/T), Pitting/Trenching Samples (PTS), Petrological Samples (PS)

1.7 Acknowledgement:

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2.0 GEOLOGY

2.1 Regional Geology

The studied area “Govindpur Copper Prospect” is located in the western part of Bastar craton and exposes basement gneisses belonging to the Bengpal Group with enclaves of older metapelites and zones of silicification. The gneisses are overlain by sandstone and quartzite equated to Penganga Group. Penganga sandstone is best exposed at 2.5 km ENE of Bothli village. It occurs as outlier trending NNW-SSE with gentle dip of 10° to 35° towards NE. At some places strike varies E-W with northerly dip of 15° to 25° . Penganga sandstone is hard, compact, medium to fine grained, brown whitish yellow or buff coloured rock due to presence of ferruginous cementing material. It mainly comprises of grit and arkose.

The area shows lineaments trending NW-SE and NNE-SSW direction. The mineralisation is essentially controlled by NS and NNE-SSW trending lineaments with silicified sheared quartz veins/reef. The silicified sheared quartz vein occurred in the form of ridges at various places in the area, viz: the ridge 1.5 km north of Sonoli, 2 km south of Motegaon and near Lawari village, are associated with copper mineralisation, malachite encrustation, pyrite, chalcopyrite and a few grains of covellite. The other sheared quartz veins which are devoid of any mineralisation have been noticed about 5 km southwest of Talodhi and about 4 km east of Nander village trending NW and at some places trend varies from NS to NE-SW with dip varying from 35° to 80° towards west.

In Govindpur and Bothli blocks, malachite encrustation is seen in quartz veins with disseminated fine specks of pyrite and chalcopyrite.

Laterite		Cenozoic
Sandstone / Quartzite with shale and conglomerate	Penganga Group	Neoproterozoic
Quartz vein- silicified zone	 Bengpal Group	Archaean
Granite gneiss- migmatite		
Banded Iron Formation		
Quartzite		

Bengal Group of Archaean age occupies part of the investigation area. It comprises of granite gneisses and migmatite with enclaves of hornblende schist, amphibolites, ultramafic and quartz/pegmatite veins. Bailadila Group of Palaeoproterozoic age (2500-2200 m.y) comprising quartzite, banded haematite quartzite with lenses of iron ores occurs as isolated outcrops in the southern part of the area. Penganga Group (equivalent) of Neoproterozoic age occupies in the northern, northeastern and south western part of the area.

Archaean gneiss forms the plain topography in the area while Penganga sandstone and quartzite forms small hillocks and ridges. Most of the area is covered by thick soil. Bengal Gneissic Group forming the basement for all the overlying lithounits, consists of granite gneiss/pegmatite with enclaves of older metamorphics (metapelites). Bengal Group of rocks have general NNW-SSE to NW-SE strike with a gentle dip towards NNE or NE direction. The foliations of Archaean gneisses and schists have NW-SE strike but NE-SW trend is also observed. The enclave of metapelites in basement gneisses trend in N35°W- S35°E with 25° to 50° dip towards SW.

The sedimentary rocks resting unconformably on the metamorphics, referable to Penganga Group, show NW-SE strike with a low dips (4° to 10°) to NE. Sandstones are medium to fine grained, hard, compact, white to buff coloured having minor secondary quartz veins at places.

2.2 Detailed geology:

The brief stratigraphy of the mapped area is as follows

Sandstone	Penganga Group	Neoproterozoic
Tourmaline vein		
Quartz vein-Silicified zone		
Amphibolite	} Bengal Group	Archaean
Quartz monzonite		
Quartz chlorite schist		
Granite gneiss		

2.2.1 Gneiss:

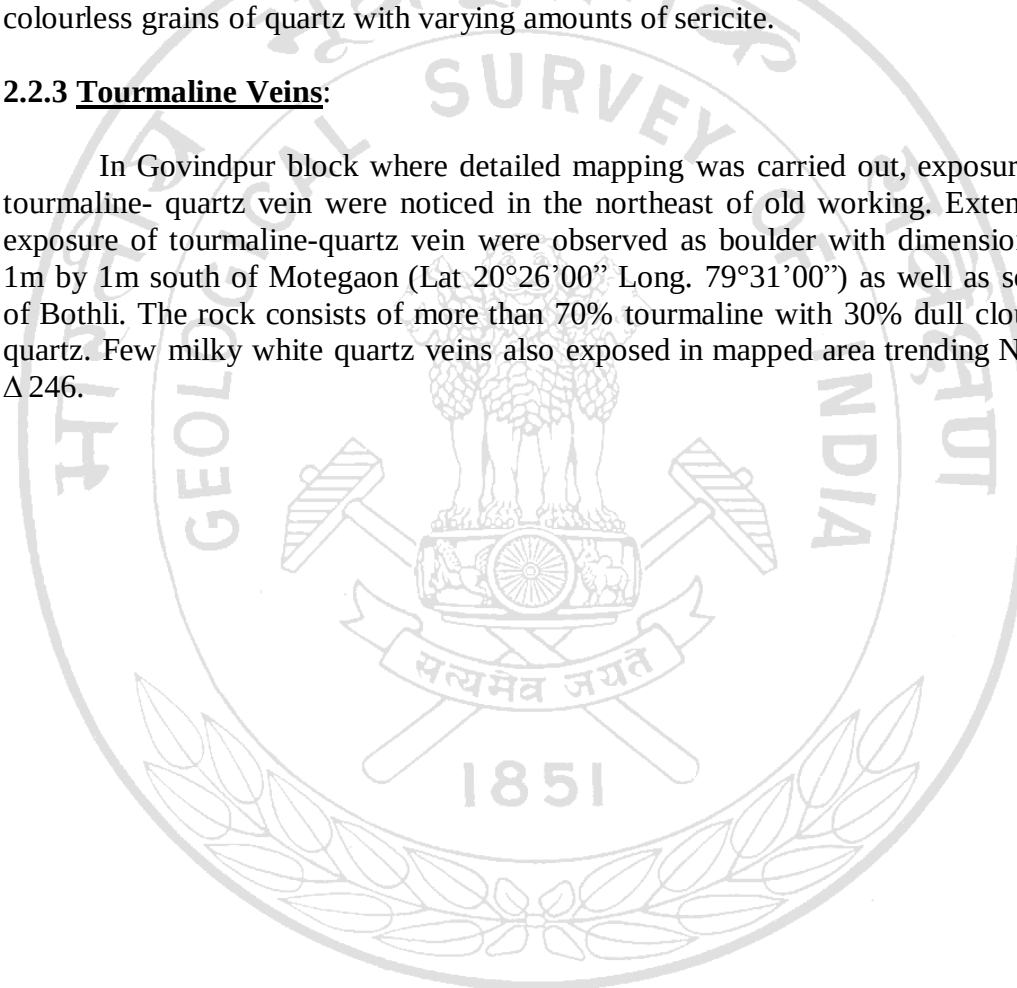
In the exposed area outcrop of gneisses are seen only in very few places, striking N20°W-S20°E, due N with subvertical dip near 1 km south of Sarangarh (Lat 29 45'00" long. 79°33'30"). It is composed of feldspar, (kaolinised) quartz, biotite and mica. The gneiss is well foliated and at places traverses by tourmaline / quartz vein.

2.2.2 Quartz Mica Shist:

It is dark greenish grey color and foliated having strike of N40°W to S40°E varying in thickness of 1m to 6m, with strike continuity of 30m. The rock shows fine grained rock is mainly composed of dark green flakes of chlorite, colourless grains of quartz with varying amounts of sericite.

2.2.3 Tourmaline Veins:

In Govindpur block where detailed mapping was carried out, exposure of tourmaline- quartz vein were noticed in the northeast of old working. Extensive exposure of tourmaline-quartz vein were observed as boulder with dimension of 1m by 1m south of Motegaon (Lat 20°26'00" Long. 79°31'00") as well as south of Bothli. The rock consists of more than 70% tourmaline with 30% dull cloured quartz. Few milky white quartz veins also exposed in mapped area trending NS at Δ 246.



3.0 Exploration

3.1 Govindpur area:

The basement rocks represented by Bengal Gneiss with metapelites, which have been intruded by pegmatite, quartz, and epidote veins, are exposed in the area. Copper mineralisation is manifested by malachite encrustation in silicified zone confined to enclave of quartz chlorite schist within Bengal gneisses which have been emplaced along N-S and NNE-SSW trending lineaments

Detailed mapping was carried out in Govindpur Block with the aid of Total Station. A base line of 600m length was laid along N-S direction. The traverse lines were laid at right angle to this base line in E-W direction at 100m interval. Pegs were put at 25m intervals along all the each traverse line of 400m length each in order to maintain the accuracy in mapping and aid in systematic bed rock and soil sampling.

The area has scanty exposures around Govindpur and Bothli as most of the area is covered by thick soil. Detailed mapping on 1:2000 was carried out in Govindpur area. Quartz vein with malachite intruding metapelites (Quartz mica schist) was mapped in Govindpur old working area.

The main old copper mine exposed in southeast of Govindpur comprising encrustation of malachite, cuprite and chalcocite are observed in silicified shear zone intruded within Bengal gneisses of archaean age. Apart from main old mine pit, two small circular shafts observed northeast of the main pit, having 4m diameter and depth upto 6m, comprising of minor amount of malachite. In order to find out thickness and extension of mineralized zone four trial pits were opened in and around main old working pit during the field season 2007-08 and it is proved to be non mineralized .

The mineralized quartz reef trending N-S having a 155m length and 4 to 7m width, is broadly folded with shallow plunge towards northeast.

A few small isolated outcrop of quartz tourmaline bands are also mapped in around Govindpur old working

The basement rocks represented by Bengal Gneiss with metapelites, which have been intruded by pegmatite, quartz, and epidote veins, are exposed in the area. Copper mineralization is manifested by malachite encrustation in silicified zone confined to enclave of metapelite within Bengal gneisses which have been emplaced along N-S and NNE-SSW trending lineaments.

Quartz reefs with sulphides intruding sheared silicified quartz mica schist was mapped in

Govindpur old working area, having N-S trend and exposed for a strike length for 155m with 4 to 5m width and is located about 25m west of main pit at Govindpur. The mineralized quartz reef trending N-S is broadly folded with low plunge of 120 to 150 towards NE

A chain of small and isolated old working pits with a huge dump material were located at Govindpur and Bothli area.

3.2 Bothli area:

Detailed mapping was carried out in Bothli Block with the aid of Total Station. A base line of 750m length was laid along N-S direction. The traverse lines were laid at right angle to this base line in E-W direction at 100m interval. Pegs were put at 25m intervals along all the each traverse line of 450m length each in order to maintain the accuracy in mapping and aid in systematic bed rock and soil sampling.

An old copper mine is located about 1.8km east of village Bothli, having length upto 200m and width varies from 1to 10m, trending N-S direction. An occurrence of silicified sheared quartzite has been noticed to the south and north of the pit, which mainly composed of malachite associated with iron oxides. A few grains of pyrite, chalcopyrite and covellite have also been noticed in silicified sheared zone exposed in a pit near 'T0' point while carrying out detailed mapping of Bothli area (plates II).

Another old working near Aregaon Chak is noticed in the area SSW of main pit in Bothli which trends NNE to NS and has a width of 1 to 4.5m. The two pits observed have shown encrustations of malachite associated with silicified sheared zone, which are highly oxidised, silicified and kaolinised.

Sulphides in the form of disseminations, stringers and veins are noticed at Bothli in the silicified zone. The main sulphides noticed in sheared silicified zone are chalcopyrite, bornite and pyrite.

Both soil and bed rock samples were collected in grid pattern (25 x100m) in Govindpur and Bothli areas. The samples are dried and meshed in 100 mesh size. The analytical values of the soil samples range from 20ppm to 0.10% Cu, <10-125 ppm for Pb, 20-500 ppm for Zn and for bed rock samples from 0.77-3.15% Cu, 10-0.18%for Pb, <10-100ppmfor Zn.

4.0 Mineralization

Mineralisation is noticed in general along the NNW-SSE trending shear zone, which consists of both metallic and non-metallic minerals. The sulphide mineralization is associated with silicified quartz reef/veins and essentially consists of chalcopyrite, pyrite, chalcocite, covellite and malachite.

The sulphide mineralisation occur mainly in the form of dissemination, fracture fillings and specks in highly fractured and silicified quartz veins, at places also occur as discrete grains in host rock.

The mineralisation is essentially structurally controlled confined to shear opening and fracture. The mineralisation appears to be hydrothermal in origin as indicated by wall rock alteration and nature of its occurrences within the host rock.

The mineralisation in the area may be attributed to hydrothermal origin related to the younger granite intrusives into the basement in the vicinity, which is evidenced by numerous pink felspar and quartz veins within the metapelites.

The Cu mineralisation in this area displays a distinct structural (fracture planes) and lithological control (altered metapelites) suggesting an epigenetic origin.

The sulphide crustification and comb structures noticed along fracture planes of silicified quartz veins and metapelites rock also suggests an epithermal type of mineralisation.

5.0 PETROGRAPHY

5.1 Quartz-chlorite mylonite:

This is a fine to medium grained rock with an assemblage of quartz, chlorite, sericite and muscovite. Lenses/ribbons of quartz are observed and developed due to high shear strain (Fig.-1). At some places quartz grains are crushed into fine aggregates. The development of granulation and recrystallization of quartz is recorded due to intense shearing (Fig.-2). Quartz and chlorite is present alternately with sericitic masses. Kinked muscovite is also noticed in this rock (Fig.-3). Zircon is occurs as an accessory mineral and oriented along the shearing.

5.2 Quartz-fibrolite mylonite:

This is a fine to medium grained rock with an assemblage of quartz, fibrolite, sericite, chlorite with less muscovite. Lenses/ribbons of quartz are observed and developed due to intense shearing (Fig.-4). Mylonitic and cataclastic texture are recorded in this rock. Quartz and fibrolite are present alternately. Fibrolite is replaced by muscovite at places (Fig.-5). Zircon, apatite are occurs as accessory minerals and oriented along the shearing.

5.3 Garnetiferous quartz monzonite:

This is medium grained rock exhibiting equigranular, hypidiomorphic texture (Fig.-6) composed of plagioclase feldspar = 45% + orthoclase = 37% + quartz 13% + garnet = 3% + (chlorite+biotite+muscovite) = 2% (by modal analysis). Garnet is deformed and altered to chlorite and biotite along the fractures. Untwined Orthoclase and conspicuous lamellar twinning in plagioclase is observed. Distorted and displaced twin lamellae along the micro fracture are noticed in plagioclase indicating the rock is subjected to intense deformation (Fig.-7). At some places recrystallized quartz are recorded.

5.4 Amphibolite:

Medium grained rock consist predominantly anthophyllite = 39% + hornblende = 17% + actinolite = 11% + calcic plagioclase = 29% + (sphene+magnetite+ilmenite) = 2 % (by modal analysis). Anthophyllite is observed and showing parallel extinction. Hornblende is green coloured, subhedral to euhedral nature. Hornblende is slightly altered to chlorite along the grain boundary. Calcic plagioclase is sausalitised completely to chlorite, epidote and sericite. At some places quartz is released from sausalitised feldspar and filled the cracks of amphibole (Fig.-8). Sphene, magnetite and ilmenite are occurs as accessory minerals.

5.5 Vein quartz:

They are mainly composed of quartz of different generations with sericite, chlorite and fine flakes of muscovite. Some of the quartz veins show euhedral, hexagonal to pseudo-hexagonal zoned quartz (Fig.-9). The zoning in quartz suggests that quartz precipitated from a polygenic silica-rich hydrothermal

fluid. Silica-rich hydrothermal fluid carries the sulphide mineralization (Pati, 2007). Some quartz is elongated and showing strained undulose extinction. At places, quartz crushed into fine aggregates is noticed (Fig.-10). All quartz grains are having abundance of submicron size inclusions. They show intense crushing, silicification, mylonitization and cataclastic deformation due to brittle ductile shearing. Post shearing quartz veins of different generations are also recorded.



6.0 Ore Petrography

In the study area, sulphide mineralization is hosted by hydrothermal quartz veins. Mineralization occurs as disseminated specks, veinlets comprising of pyrite, chalcopyrite, covellite and paxite.

6.1 Pyrite (FeS_2):

It occurs as veins, cavity fillings as well as in disseminations (Fig.-11) with euhedral outline and typical pitted nature due to its high hardness. It is light yellow colour with high reflectance and distinct isotropism.

6.2 Chalcopyrite (CuFeS_2):

It occurs as a minor phase in quartz vein. It is medium grained with pure yellow colour and shows weak anisotropism in greenish grey tints.

6.3 Covellite (CuS):

It occurs as lamellae of pure blue coloured (Fig.-12), shows strong pleochroism and extreme anisotropism in orange brown (Fig.-13).

6.4 Paxite (CuAs_2):

It is fine grained, subidiomorphic, yellow-white in colour with high reflectance and occurs as discrete grains with typical twin lamellae and distinct anisotropism.

6.5 Paragenesis:

The ore petrographic studies of the mineralized zones of Govindpur and Bothli blocks shows prominent sulphide mineralization viz., pyrite, chalcopyrite, covellite and paxite along with hematite representing the oxide phase. From the ore petrographic study it is very much evident that at least three generations of mineralization are present in the area. The early phase is marked by chalcopyrite, covellite and paxite in decreasing abundance followed by distinct pyrite in the second phase. Hematite occurs as discrete grains filling cavities and veins forms last phase of mineralization. Chalcopyrite and covellite shows mutual boundary texture indicating their coeval nature. However, paxite occurs as discrete grains with typical twin lamellae and distinct anisotropism. Pyrite shows prominent replacement texture, where it is replacing chalcopyrite. Undigested chalcopyrite is recorded within the pyrite. All these phases of mineralization can be correlated with different stages of shearing and quartz reef/vein intrusion. First phase of mineralization is mainly associated with syn shearing quartz reef intrusion and second as well as third/last phase are associated with post shearing quartz veins.

PART-II

7.0 Geochemistry and X-RD study:

7.1 Geochemistry:

Whole rock analysis of 3 samples was received from Chemical Division, CR, Nagpur. The analysis was carried out by XRF technique. The rocks are amphibolite and mylonite.

(Table-I)

Whole rock analysis of 3 samples from Govindpur and Bothli block.

Sl. No.	Sample No.	Lithology	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	MgO	CaO	Na ₂ O	K ₂ O	MnO	TiO ₂	P ₂ O ₅	LOI
1	TL34a	Amphibolite	51.26	12.44	15.35	6.88	7.42	2.29	1.22	0.24	0.95	0.08	2.36
2	TL6a	Quartz monzonite	75.09	13.56	1.14	0.46	1.24	7.58	0.40	0.09	Trace	Trace	0.54
3	TL41a/L1	Quartz fibrolite mylonite	89.41	3.30	3.07	1.10	0.15	Trace	0.58	Trace	0.18	Trace	1.42

7.2 X-RD Study:

The X-RD studies were carried out in 10 samples. The study reveals that garnet from monzonite is almandine. Pyrite is found in quartz veins. However, no gold is identified. The X-ray diffraction analysis of samples was done at XRD lab, GSI, Faridabad. The detailed analysis of report is given below.

Sample No.	Major mineral phase	Minor mineral phase	Trace mineral phase
G3	Quartzite- α	Magnetite	Chlorite
TL6a	Quartz, Albite	Biotite	Orthoclase
TL6a/G	Almandine, Albite	Quartz low	----
TL6a/G	Quartz low	Nimmite	Magnetite, Ilmenite
TL19c	Quartz low	Nimmite	Chalcocite, Ilmenite, Haematite
TL29	Actinolite and Albite	Birnessite	Vermiculite, Sphene
TL34a	Quartz	Biotite, Nimmite	Pyrite, Sapontite
TL41B	Quartz low	Muscovite	Clinochlore, Arsenopyrite
TL41C	Quartz	---	Kaolinite, Haematite



8.0 Geophysical Study:

The layout of geophysical traverses at Govindpur copper prospect, falling in the Survey of India Toposheet No. 55P/11, along with geology is shown in Fig.-1. The N08°E- S08°W oriented 0.85 km baseline was laid during the field season 2007-08. Thus, N82°W- S82°E oriented seventeen traverses between N60 and S20 were laid at 100m separation orthogonal to baseline drawn with respect to a reference point (0/0). These traverses laid over a length of 0.8 km were used to carry out geophysical surveys at a station interval of 10m between stations W20 and E30 on either side of the baseline. An area of 0.85 sq km was covered by geophysical surveys during the field season 2007-08.

The high chargeability and low resistivity (Fig.-8) has indicated a N-S intrusive body at W10 station having a structural control between the traverses S20 and N27 with steeply dipping towards west and gentle dip towards east may indicate a quartz vein. This steeply dipping vein might not be carrying any mineralisation. However, at the eastern end, the low resistivity of the order of 200 ohm-m supported by high chargeability of the order of more than 10mV/V indicate mineralisation. This part of the mineralisation is southern end situated near the old working, which falls as a northern extension at the stations E5 to E10 along the traverse interval of N2 to N12. It is situated east of the baseline by 50m and falls as a part of in high chargeability in the category of high chargeability and low resistivity. But remaining part of the high chargeability is associated with high resistivity (Fig.-9) formations of the resistivity values more than 600ohm-m that indicate the mineralisation is highly disseminated. Although the combination of low SP and high chargeability is a good indicator for locating the copper mineralization, this particular combination (Fig.-10) has not existed. However, the low SP, high chargeability associated with low resistivity is observed around the stations E5 to E10 through the traverses N2 to N12.

All these conclusions mentioned above are based on qualitative methods. To have the depth control, which is a quantitative method, Vertical Electrical Sounding (VES) and Dipole- Dipole (Dp-Dp) are analysed. The VES (Fig.-4) at N15/E3 concluded that the chargeability increases with increase in depth, except a part of the sounding between the AB/2 spacing at 30 and 100m. Both chargeability and reaches saturation. However, both the curves again continued its increasing trend. At a separation beyond 100m the resistivity and chargeability values attain 280 ohm-m and 14mV/V respectively. This is very much in coincidence with the Fig 8, where the low resistivity of the order of 300 ohm-m and chargeability to 12mV/V along with low SP is suggested for occurrence of copper mineralisation. In addition to VES, Dp-Dp method (Fig.-5) also contributed to estimate depth in terms of levels/depth along the traverses N10 and N20. It confirmed the possible occurrence of mineralisation at 45m depth (3rd level) along both the traverses. The profiling (Fig.-11) plotted along the traverses '0', N10 and N20 also indicate that the western end of the high resistivity is associated with low chargeability compare to the eastern end which is promising as far as copper mineralisation is concerned.

9.0 Conclusions and Recommendations

Detailed geological mapping were carried out in both areas to probe mineralization at depth and its strike continuity in the area.

From the data collected by soil samples analysis in Govindpur and Bothli areas of Chandrapur District, it is concluded that the mineralization is restricted in extent without much depth persistence.

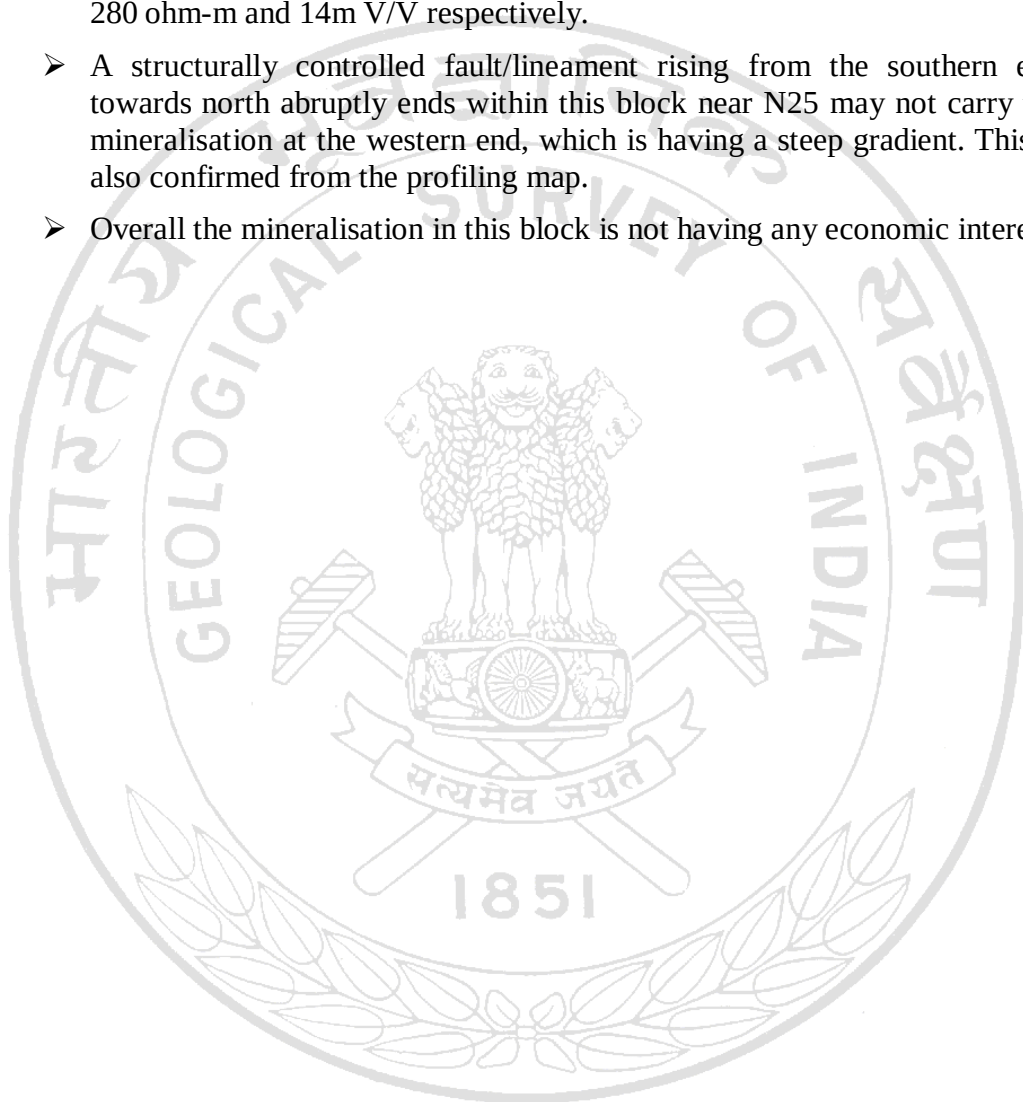
The mineralization zone in Bothli area has considerable length and width. The quality of the ore is also good. Observation of the trial pits opened in the area indicated that wall rocks are highly altered by hydrothermal agencies as is evident from silicification and Chloritisation of the rock.

West of Govindpur, an area of 2km.towards Bothli old working further needs chemical analysis of soil samples, extensive pitting and trenching to know strike continuity of copper ore horizon.

- In Govindpur and Bothli block, the main rock types present are quartz-chlorite mylonite, quartz-fibrolite mylonite, garnetiferous quartz monzonite, and amphibolites with younger mineralized sheared quartz reef/vein.
- Intense mylonitisation and brecciation in mylonite and quartz vein indicates brittle- ductile shearing occurred in the area.
- Development of lenses/ribbons, crushing, granulation and recrystallisation of quartz indicates intense shearing. The zoning in quartz in silicified quartz vein observed under microscope suggests that quartz precipitated from a polygenic silica-rich hydrothermal fluid, which is also the carrier of sulphide mineralization.
- The general sulphide mineral assemblage recorded in the area include pyrite, chalcopyrite, covellite and paxite (CuAs_2) along with hematite representing the oxide phase.
- At least three distinct phases of mineralization is recorded based on textural relationship between ore minerals and ore and gangue minerals marked by both sulphide and oxide phases.
- All these phases of mineralization can be correlated with different stages of shearing and quartz reef/vein intrusion. First phase of mineralization is mainly associated with syn shearing quartz reef intrusion and second as well as third/last phase are associated with post shearing quartz veins.
- Pervasive hydrothermal activity in the area resulted intense wall rock alteration viz, silicification, sericitisation, chloritisation, sausalitization indicating an epithermal to mesothermal nature of mineralization.

In Govindpur copper prospect, no gold mineralization is found. Sulphide minerals viz. pyrite, chalcopyrite, covellite, paxite are found but in less concentration.

- In Govindpur block the possibility of occurrence of copper mineralisation is limited to only 100 X 50sq.m, as low resistivity, high chargeability and low SP are concentrated here.
- It is also recommended that the mineralisation is limited to the traverses '0', N10 and N20.
- The VES indicate that the chargeability increases while probing deeper. But at a particular combination of probe spacing both the resistivity and chargeability values reach saturation point at AB/2 is between 30 and 100m. On further increase in AB/2 spacing, both the parameters increase to 280 ohm-m and 14m V/V respectively.
- A structurally controlled fault/lineament rising from the southern end towards north abruptly ends within this block near N25 may not carry the mineralisation at the western end, which is having a steep gradient. This is also confirmed from the profiling map.
- Overall the mineralisation in this block is not having any economic interest.



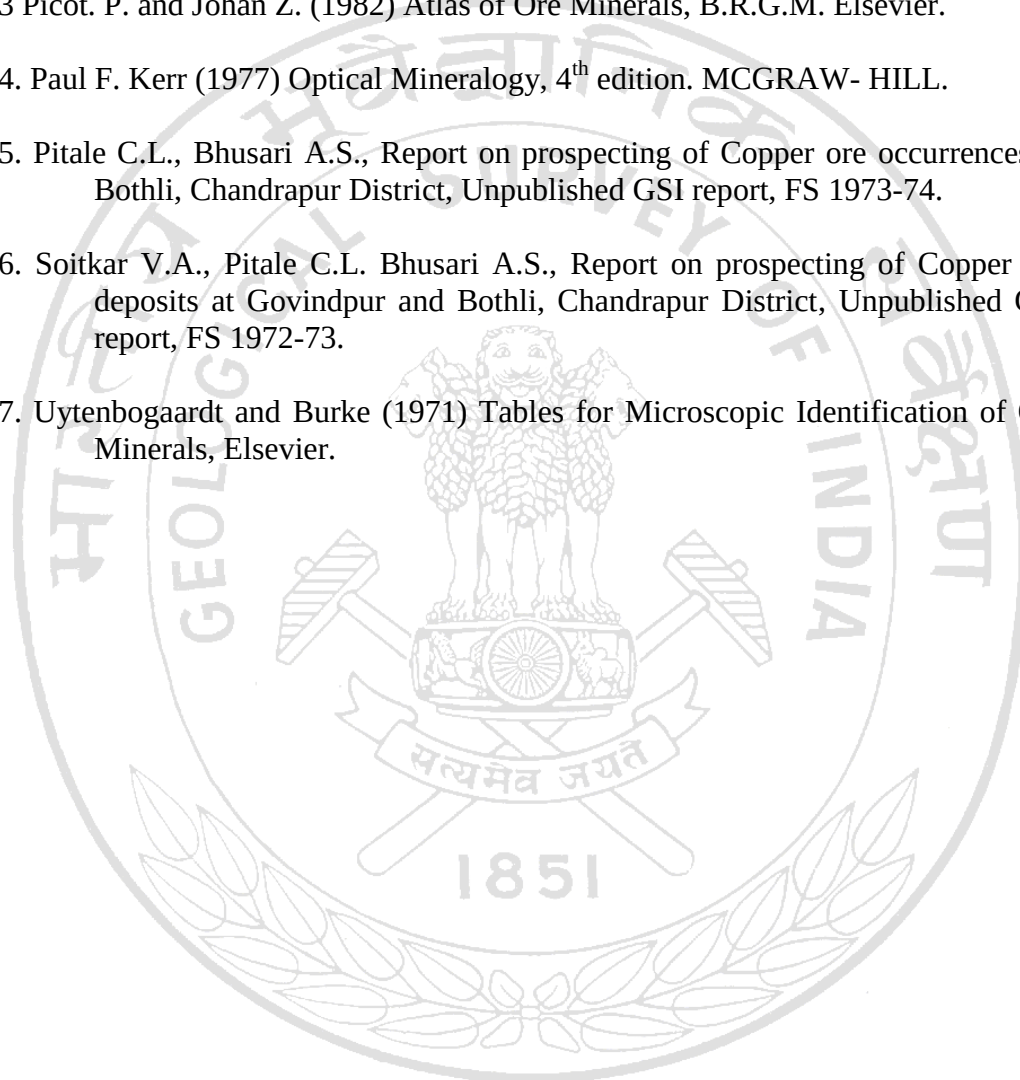
Locality Index:

Location	Latitude			Longitude			Toposheet No.
	Deg	Min	Sec	Deg	Min	Sec	
Bothli	20	28	00	79	31	00	55P/11
Govindpur	20	29	00	79	37	00	55P/11
Kitari	20	30	00	79	39	00	55P/11
Motegaon	22	26	00	79	31	00	55P/11
Taka	20	40	00	79	24	00	55P/11
Neri	20	28	00	79	27	00	55P/11
Talodi	20	26	00	79	40	00	55P/11
Lawari	20	30	00	79	32	00	55P/11
Jabhulghata	20	32	00	79	28	00	55P/11
Metapur	20	33	00	79	27	00	55P/11
Yeonoli	20	28	12	79	35	62	55P/11
Lohara	20	30	00	79	31	00	55P/11
Sirpur	20	28	17	79	30	00	55P/11
Palasgaon	20	28	00	79	41	00	55P/11



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ANNEXURE-I

**CHEMICAL ANALYSIS OF SOIL SAMPLES, GOVINDPUR AREA,
CHANDRAPUR DISTRICT**

Sr. No.	Ref.No.	Cu	Pb	Zn	Co	Ni	Ag	As	Au	All values in ppm , Au in ppb
1	SS-1	0.10%	30	50	20	40	<1	10	<100	
2	SS-2	350	30	175	20	60	<1	10	<100	
3	SS-3	75	20	50	20	60	<1	20	<100	
4	SS-4	125	75	200	60	60	<1	10	<100	
5	SS-5	60	50	125	60	75	<1	20	<100	
6	SS-6	100	125	100	100	225	<1	10	<100	
7	SS-9	30	40	50	30	40	<1	10	<100	
8	SS-10	30	30	40	10	30	<1	10	<100	
9	SS-11	40	40	75	20	40	<1	2	<100	
10	SS-12	40	20	100	10	40	<1	<1	<100	
11	SS-13	40	30	125	20	50	<1	<1	<100	
12	SS-14	50	30	250	40	60	<1	4	<100	
13	SS-15	40	20	175	20	50	<1	4	<100	
14	SS-16	30	30	100	30	60	<1	<1	<100	
15	SS-17	40	30	175	30	60	<1	10	<100	
16	SS-18	20	40	60	20	40	<1	4	<100	
17	SS-19	40	30	60	30	60	<1	4	<100	
18	SS-20	30	30	50	10	40	<1	<1	<100	
19	SS-21	40	30	75	20	60	<1	<1	<100	
20	SS-22	20	30	40	20	50	<1	<1	<100	
21	SS-23	50	60	75	75	100	<1	2	<100	
22	SS-24	40	40	75	20	75	<1	<1	<100	
23	SS-25	40	40	75	20	75	<1	4	<100	
24	SS-26	60	60	100	50	100	<1	4	<100	
25	SS-27	40	40	60	20	60	<1	4	<100	
26	SS-28	40	75	100	30	60	<1	4	<100	
27	SS-29	40	40	60	30	50	<1	10	<100	
28	SS-30	150	30	150	<10	20	<1	4	<100	
29	SS-31	200	100	60	150	150	<1	<1	<100	
30	SS-32	60	60	60	150	200	<1	2	<100	
31	SS-33	30	20	50	20	50	<1	10	<100	
32	SS-34	60	75	175	125	100	<1	20	<100	
33	SS-35	20	20	60	10	30	<1	30	<100	
34	SS-36	20	20	200	<10	10	<1	<1	<100	
35	SS-37	50	30	100	20	75	<1	<1	<100	
36	SS-38	30	10	40	10	40	<1	<1	<100	
37	SS-40	50	30	100	30	100	<1	10	<100	
38	SS-41	40	30	75	30	60	<1	10	<100	
39	SS-42	40	30	100	40	75	<1	<1	<100	
40	SS-43	40	30	100	30	60	<1	4	<100	
41	SS-44	30	50	100	50	60	<1	10	<100	
42	SS-45	30	20	100	20	40	<1	10	<100	
43	SS-46	60	20	40	20	40	<1	10	<100	
44	SS-47	50	20	100	30	60	<1	4	<100	
45	SS-48	20	10	60	20	20	<1	10	<100	
46	SS-49	75	150	75	150	100	<1	20	<100	
47	SS-50	10	<10	500	<10	<10	<1	<1	<100	
48	SS-51	30	20	60	20	40	<1	4	<100	

Sr. No.	Ref.No.	Cu	Pb	Zn	Co	Ni	Ag	As	Au	
49	SS-52	20	10	60	20	40	<1	<1	<100	
50	E-1	475	30	40	40	40	<1	10	<100	
51	E-2	150	20	60	30	50	<1	10	<100	
52	E-3	75	10	40	30	40	<1	4	<100	
53	E-4	50	10	20	20	30	<1	4	<100	
54	E-5	40	10	30	20	40	<1	10	<100	
55	E-6	40	10	100	20	40	<1	<1	<100	
56	E-7	30	10	40	10	30	<1	4	<100	
57	E-8	50	20	60	20	50	<1	<1	<100	
58	E-9	50	10	40	20	40	<1	<1	<100	
59	E-10	125	30	50	40	75	<1	4	<100	
60	E-11	40	20	20	20	40	<1	4	<100	
61	E-12	50	10	50	20	40	<1	4	<100	
62	N2	200	20	30	50	60	<1	<1	<100	
63	N3	30	10	30	20	50	<1	10	<100	
64	N3E1	30	20	40	20	50	<1	4	<100	
65	N3E2	20	20	20	10	30	<1	4	<100	
66	N2E1	50	30	30	30	40	<1	<1	<100	
67	N2E2	75	30	40	30	50	<1	10	<100	
68	N2E3	20	20	20	20	30	<1	<1	<100	
69	N2E4	50	30	30	20	40	<1	20	<100	
70	N2E5	30	30	30	20	40	<1	<1	<100	
71	N2E6	20	30	50	10	30	<1	10	<100	
72	N2E7	30	30	100	20	40	<1	10	<100	
73	N2E8	40	50	60	30	50	<1	20	<100	
74	N2E9	20	30	60	20	30	<1	10	<100	
75	N2E10	20	30	20	20	30	<1	4	<100	
76	N2E11	20	30	75	20	30	<1	10	<100	
77	N2E12	30	30	125	20	40	<1	10	<100	
78	N1W1	310	<10	280	30	45	<5	10	110	All values in ppm, Au in ppb
79	N1W2	225	10	190	15	20	<5	20	120	
80	N1W4	95	<10	395	20	40	<5	10	<100	
81	N1	30	<10	305	20	40	<5	2	120	
82	N3E5	90	<10	220	10	30	<5	2	100	
83	N3E7	75	<10	370	10	50	<5	4	120	
84	N3E8	35	<10	390	10	60	<5	10	140	
85	N3E9	50	<10	330	15	40	<5	20	120	
86	N3E6	25	<10	640	15	50	<5	20	140	
87	N3E3	30	<10	215	30	<10	<5	2	150	
88	N3E4	50	25	295	10	40	<5	10	170	
89	N3W3	915	<10	430	10	70	<5	10	140	
90	N1E1	65	<10	60	30	60	<5	4	190	
91	N1E2	50	<10	50	25	40	<5	<1	190	
92	N1E3	60	15	75	30	70	<5	2	230	
93	N1E4	60	15	70	45	65	<5	4	210	
94	N1E5	25	<10	70	20	35	<5	<1	210	
95	N1E6	30	10	50	<10	40	<5	2	230	
96	N1E7	20	10	45	30	40	<5	4	230	
97	N1E8	25	10	100	10	40	<5	4	210	
98	N1E9	20	<10	45	40	40	<5	<1	230	
99	N1E10	25	10	55	25	45	<5	4	<100	

Sr. No.	Ref.No.	Cu	Pb	Zn	Co	Ni	Ag	As	Au	
100	N1E11	30	15	110	20	40	5	4	<100	
101	N1E12	30	10	55	25	40	<5	4	130	
102	N3E10	25	10	70	25	45	5	<1	140	
103	N3E11	60	10	70	20	50	7	10	120	
104	N3E12	40	<10	60	25	45	5	10	120	
105	N1W3	490	15	35	75	60	5	4	120	
106	N3W1	50	15	45	25	60	7	4	130	
107	N3W2	50	10	50	55	65	5	<1	130	
108	N3W4	45	30	50	10	60	6	<1	100	
109	N2W3	75	15	40	55	60	60	<1	<100	
110	S1	460	<10	30	20	30	35	<1	<100	
111	S1W1	460	10	45	20	60	30	<1	<100	
112	S1W2	360	30	45	25	60	35	<1	<100	
113	S1W3	395	<10	35	20	35	50	1	<100	
114	S1W4	370	10	25	10	30	65	<1	<100	
115	S2	45	10	30	10	30	50	<1	<100	
116	S2W1	70	10	35	40	70	20	<1	<100	
117	S2E1	30	10	20	40	15	20	<1	<100	
118	S2E2	285	<10	380	65	10	25	10	150	
119	S2E3	365	<10	40	40	35	60	<1	<100	
120	S2E4	370	<10	20	30	20	20	<1	<100	
121	S2E5	255	<10	35	45	25	70	<1	<100	
122	S2E6	85	<10	30	30	60	95	<1	<100	
123	S2E7	90	<10	65	25	45	90	<1	<100	
124	S2E8	40	15	35	15	25	70	<1	<100	
125	S2E9	60	10	45	15	35	30	<1	<100	
126	S2E10	65	<10	120	20	60	25	<1	<100	
127	S2E11	45	10	40	10	45	35	<1	<100	
128	S2E12	35	<10	30	30	20	35	<1	<100	
129	S1E1	280	10	30	50	75	30	2	<100	
130	S1E3	185	<10	20	30	45	35	<1	<100	
131	S1E4	145	<10	20	55	40	35	<1	<100	
132	S1E5	125	<10	30	60	40	50	<1	<100	
133	S1E7	70	<10	35	20	40	20	<1	<100	
134	S1E8	45	<10	35	30	15	25	1	<100	
135	S1E9	35	10	35	20	30	10	<1	<100	
136	S1E11	70	<10	40	40	15	10	<1	<100	
137	S1E12	55	15	40	25	20	25	1	<100	
138	S3E4	20	100	150	30	15	25	<1	<100	
139	S3E5	40	300	460	30	35	20	<1	125	
140	S3E6	60	800	630	40	25	25	<1	150	
141	S3E7	55	95	215	30	35	30	<1	100	
142	S3E8	70	50	230	20	45	30	<1	<100	
143	S3E10	75	30	80	30	50	20	2	<100	
144	S3E11	65	40	90	50	100	20	<1	125	
145	S.5	0.64%	15	25	35	50	20	<1	300	
146	S.5W1	0.22%	<10	35	30	40	25	<1	175	
147	S.5W2	0.13%	10	20	30	20	30	2	200	
148	S.5W3	0.38%	45	30	40	30	25	<1	175	
149	S.5W4	0.10%	10	20	25	25	<5	10	<100	
150	S.5E1	0.38%	20	20	25	30	<5	10	425	

All values
in ppm and
Au in ppb

Sr. No.	Ref.No.	Cu	Pb	Zn	Co	Ni	Ag	As	Au	
151	S.5E2	90	15	25	30	30	<5	1	<100	
152	S2W1	970	<10	35	30	35	<5	2	<100	
153	S2W2	85	<10	25	30	30	15	4	<100	
154	S2W3	75	<10	25	30	30	15	<1	<100	
155	S2W4	60	10	30	10	30	10	<1	<100	
156	S3	25	10	20	15	15	10	<1	<100	
157	S3E1	50	10	35	15	35	<5	2	<100	
158	S3E2	30	15	35	10	20	<5	1	<100	
159	S3E3	35	50	60	10	35	<5	<1	<100	
160	S3E9	95	30	120	30	120	<5	<1	<100	
161	S3E12	60	35	140	30	110	10	2	<100	
162	S3W1	50	10	140	20	30	10	<1	<100	
163	S3W2	50	<10	30	30	40	15	10	<100	
164	S3W3	45	<10	40	20	45	25	1	100	
165	S3W4	50	<10	35	50	25	35	2	100	
166	S1E2	330	<10	60	50	80	65	10	100	
167	S1E6	100	<10	25	25	30	60	2	<100	
168	S1E10	60	<10	25	20	30	60	<1	<100	
169	N2W2	110	<10	50	25	45	50	2	100	
170	N2W4	60	10	30	15	50	30	1	125	
171	S1	40	20	40	30	60	<1	<5	<100	
172	S2	30	10	30	10	40	<1	<5	<100	
173	S3	40	20	60	30	75	<1	<5	<100	
174	S4	50	20	60	30	75	<1	<5	<100	
175	MT-1	40	10	30	20	40	<1	<5	<100	
176	MT-2	40	60	30	30	50	<1	<5	<100	
177	G-5	40	50	60	40	75	<1	<5	<100	
178	14658/G-	45	10	45	35	40	<1	<5	<100	
179	G-16	45	20	65	20	45	<1	<5	<100	
180	G-18	25	10	35	20	25	<1	<5	<100	
181	G-19	40	50	60	30	75	<1	<5	<100	
182	G-20	40	20	35	15	35	<1	<5	<100	
183	G-21	40	50	75	30	75	<1	<5	<100	
184	G-22	30	20	25	20	40	<1	<5	<100	
185	G-23	40	50	60	40	75	<1	<5	<100	
186	G-24	40	45	35	15	55	<1	<5	<100	
187	G-25	30	30	50	20	60	<1	<5	<100	
188	G-27	40	45	20	20	45	<1	<5	<100	
189	G-29	30	50	50	40	75	<1	<5	<100	
190	G-30	40	30	25	10	45	<1	<5	<100	
191	G-32	40	25	25	45	20	<1	<5	<100	
192	G-33	40	35	20	20	60	<1	<5	<100	
193	G-35	25	30	25	35	30	<1	<5	<100	
194	G-36	30	20	40	30	60	<1	<5	<100	
195	G-38	60	30	50	50	100	<1	<5	<100	
196	G-39	55	15	40	35	45	<1	<5	<100	
197	G-40	30	20	50	30	75	<1	<5	<100	
198	G-51	40	30	50	20	75	<1	<5	<100	
199	G-52	40	40	50	30	60	<1	<5	<100	
200	G-53	30	40	60	30	60	<1	<5	<100	
201	G-54	40	40	60	30	60	<1	<5	<100	

All values
in ppm and
Au in ppb

Sr. No.	Ref.No.	Cu	Pb	Zn	Co	Ni	Ag	As	Au	
202	G-55	30	40	60	20	50	<1	<5	<100	
203	G-56	40	40	50	30	60	<1	<5	<100	
204	G-57	40	60	75	50	75	<1	<5	<100	
205	G-58	40	40	75	30	60	<1	<5	<100	
206	G-59	20	40	60	20	60	<1	<5	<100	
207	G-60	50	50	50	20	50	<1	<5	<100	
208	G-61	60	40	75	40	75	<1	<5	<100	
209	G-62	50	60	60	60	100	<1	<5	<100	
210	G-63	75	75	75	40	100	<1	<5	<100	
211	G-64	50	50	75	30	60	<1	<5	<100	
212	G-65	60	100	75	100	125	<1	<5	<100	
213	G-66	40	60	60	5	300	<1	<5	<100	
214	G-67	40	50	75	30	200	<1	<5	<100	
215	G-68	30	50	50	20	75	<1	<5	<100	
216	G-69	30	40	60	30	125	<1	<5	<100	
217	G-70	50	60	75	6	100	<1	<5	<100	
218	G-71	30	50	75	20	100	<1	<5	<100	
219	G-72	30	40	60	20	75	<1	<5	<100	
220	G-73	40	50	75	30	75	<1	<5	<100	
221	G-74	30	40	50	20	60	<1	<5	<100	
222	G-75	50	75	75	100	150	<1	<5	<100	
223	G-76	30	50	60	20	50	<1	<5	<100	
224	G-77	30	50	60	20	60	<1	<5	<100	
225	G-78	30	50	60	30	50	<1	<5	<100	
226	G-79	45	40	40	35	100	<1	<5	<100	
227	G-80	70	40	60	60	80	<1	<5	<100	
228	G-81	80	130	60	100	110	<1	<5	<100	
229	G-82	20	40	40	10	4	<1	<5	<100	
230	G-83	30	40	50	20	50	<1	<5	<100	
231	G-84	50	50	60	30	100	<1	<5	<100	
232	G-85	30	40	50	20	50	<1	<5	<100	
233	G-86	50	50	100	40	60	<1	<5	<100	
234	G-87	30	40	60	30	50	<1	<5	<100	All values in ppm and Au in ppb
235	G-88	60	75	75	75	100	<1	<5	<100	
236	G-89	50	40	60	40	750	<1	<5	<100	
237	G-90	40	60	60	40	750	<1	<5	<100	
238	G-91	40	50	75	30	60	<1	<5	<100	
239	G-92	20	40	50	20	50	<1	<5	<100	
240	G-93	30	40	60	20	50	<1	<5	<100	
241	G-94	20	40	50	20	30	<1	<5	<100	
242	G-95	50	40	60	20	60	<1	<5	<100	
243	G-96	30	30	50	20	50	<1	<5	<100	
244	G-97	30	40	60	20	50	<1	<5	<100	
245	G-98	60	100	60	75	100	<1	<5	<100	
246	G-99	40	40	60	30	50	<1	<5	<100	
247	G-100	30	40	75	20	50	<1	<5	<100	
248	G-101	40	50	75	40	60	<1	<5	<100	
249	G-102	50	40	75	30	60	<1	<5	<100	
250	G-103	35	<10	35	10	55	<1	<5	<100	
251	G-104	60	105	42	150	120	<1	<5	<100	
252	G-105	60	15	35	15	45	<1	<5	<100	

Sr. No.	Ref.No.	Cu	Pb	Zn	Co	Ni	Ag	As	Au	
253	G-106	35	15	30	45	55	<1	<5	<100	
254	G-107	35	<10	40	20	60	<1	<5	<100	
255	G-108	20	10	35	40	70	<1	<5	<100	
256	G-109	45	<10	30	10	75	<1	<5	<100	
257	G-110	40	<10	40	20	55	<1	<5	<100	
258	G-111	40	<10	30	25	55	<1	<5	<100	
259	G-112	250	<10	30	40	80	<1	<5	<100	
260	G-113	45	10	45	20	65	<1	<5	<100	
261	G-114	40	10	40	25	50	<1	<5	<100	
262	G-115	45	10	45	25	70	<1	<5	<100	
263	G-116	15	<10	20	30	45	<1	<5	<100	
264	G-117	35	10	40	25	55	<1	<5	<100	
265	G-118	50	40	40	95	95	<1	<5	<100	
266	G-119	20	<10	25	10	30	<1	<5	<100	
267	G-120	35	10	30	25	40	<1	<5	<100	
268	G-121	40	10	40	25	40	<1	<5	<100	
269	G-122	40	<10	40	25	40	<1	<5	<100	
270	G-123	20	<10	20	10	30	<1	<5	<100	
271	G-124	30	10	35	10	35	<1	<5	<100	
272	G-125	30	<10	35	15	30	<1	<5	<100	
273	G-126	35	<10	40	15	30	<1	<5	<100	
274	G-127	20	10	20	15	20	<1	<5	<100	
275	G-128	20	10	20	20	20	<1	<5	<100	
276	G-129	20	10	25	25	20	<1	<5	<100	
277	G-130	50	75	40	95	135	<1	<5	<100	
278	G-131	40	25	30	35	55	<1	<5	<100	
279	G-132	50	15	50	35	50	<1	<5	<100	
280	G-133	90	45	25	35	40	<1	<5	<100	
281	G-134	90	10	25	42	50	<1	<5	<100	
282	G-135	55	10	25	25	35	<1	<5	<100	
283	G-136	25	15	15	25	30	<1	<5	<100	
284	G-137	20	10	15	20	25	<1	<5	<100	
285	G-138	50	10	25	20	55	<1	<5	<100	
286	G-139	170	<10	15	25	40	<1	<5	<100	All values in ppm and Au in ppb
287	G-140	25	<10	20	15	35	<1	<5	<100	
288	G-141	15	<10	15	<10	20	<1	<5	<100	
289	G-142	20	10	15	25	30	<1	<5	<100	
290	G-143	25	15	25	15	40	<1	<5	<100	
291	G-144	30	<10	20	10	50	<1	<5	<100	
292	G-145	30	10	20	20	45	<1	<5	<100	
293	G-146	45	10	30	15	65	<1	<5	<100	
294	G-147	45	10	35	25	65	<1	<5	<100	
295	G-148	40	<10	30	10	60	<1	<5	<100	
296	G-149	35	10	30	20	55	<1	<5	<100	
297	G-150	15	<10	20	<10	35	<1	<5	<100	
298	G-151	35	20	35	25	55	<1	<5	<100	
299	G-152	50	<10	40	20	45	<1	<5	<100	
300	G-153	35	130	30	30	50	<1	<5	<100	
301	G-154	30	85	30	15	45	<1	<5	<100	
302	G-155	35	70	35	25	50	<1	<5	<100	
303	G-156	30	75	30	20	45	<1	<5	<100	

Sr. No.	Ref.No.	Cu	Pb	Zn	Co	Ni	Ag	As	Au	
304	G-157	25	240	30	10	35	<1	<5	<100	
305	G-158	25	120	30	<10	35	<1	<5	<100	
306	G-159	30	120	35	10	40	<1	<5	<100	
307	G-160	30	145	35	20	50	<1	<5	<100	
308	G-161	35	<10	40	15	30	<1	<5	<100	
309	G-162	30	10	35	25	45	<1	<5	<100	
310	G-163	40	<10	50	10	55	<1	<5	<100	
311	G-191	58	15	123	25	63	<1	<5	100	
312	G-192	56	14	83	25	62	<1	<5	100	
313	G-195	67	13	100	27	69	<1	<5	140	
314	G-196	31	10	56	15	30	<1	<5	100	
315	G-197	40	9	55	20	34	<1	<5	<100	
316	G-199	47	7	50	24	50	<1	<5	110	
317	G-201	66	30	78	30	71	<1	<5	110	
318	G-202	102	38	53	31	51	<1	<5	140	
319	G-203	109	40	57	25	53	<1	<5	<100	
320	G-204	108	36	49	22	44	<1	<5	<100	
321	G-205	105	9	59	16	76	<1	<5	<100	
322	G-206	83	11	51	16	48	<1	<5	<100	
323	G-207	37	11	45	14	45	<1	<5	<100	
324	G-208	44	19	68	37	51	<1	<5	<100	All value in ppm Au in ppb
325	G-209	40	29	61	18	47	<1	<5	<100	
326	G-210	64	30	65	41	67	<1	<5	<100	
327	G-211	55	31	67	35	39	<1	<5	<100	
328	G-212	46	30	61	23	31	<1	<5	<100	
329	G-213	47	42	57	60	72	<1	<5	<100	
330	G-214	40	22	47	15	50	<1	<5	<100	
331	G-215	53	24	82	15	36	<1	<5	<100	
332	G-216	24	23	41	11	19	<1	<5	<100	
333	G-217	46	23	54	15	54	<1	<5	<100	
334	G-218	49	28	57	9	56	<1	<5	<100	
335	G-219	38	13	43	11	44	<1	<5	<100	
336	G-220	31	12	38	10	35	<1	<5	<100	

ANNEXURE-II										
CHEMICAL ANALYSIS OF BED ROCK SAMPLES, GOVINDPUR AREA, CHANDRAPUR										
Sr. No.	Ref.No.	Cu	Pb	Zn	Co	Ni	Ag	As	Au	Lithology
1	BRS-53	650	<10	75	30	40	<1	<1	<100	Silicified sheared quartzite
2	BRS-A2	575	<10	350	40	20	<1	4	<100	Silicified sheared quartzite
3	BRS-A3	1.62&	<10	100	10	10	<1	10	<100	Silicified sheared quartzite
4	L1	0.77%	30	0.18%	<10	<10	<1	<1	<100	Silicified sheared quartzite
5	L2	3.15%	20	0.10%	<10	<10	<1	<1	<100	Silicified sheared quartzite
6	W1	10	<10	315	10	30	<5	<1	<100	Quartz chlorite schist
7	W2	480	<10	750	15	20	<5	20	<100	Quartz chlorite schist
8	W3	955	20	50	15	40	7	20	130	Quartz chlorite schist
9	W4	15	60	270	30	50	<5	20	200	Quartz chlorite schist
10	KGC1	20	<10	40	50	<10	25	<1	175	Tourmaline quartz vein
11	KGC2	20	<10	30	70	<10	20	2	100	Quartz vein
12	KGC3	10	10	40	80	<10	<5	<1	100	Quartz vein
13	RCG1	10	30	260	60	<10	<5	<1	<100	Quartz chlorite schist
14	RCG2	10	10	65	140	<10	<5	<1	100	Smoky quartz vein
15	RCG3	10	25	190	130	<10	<5	<1	100	Quartz chlorite schist
16	G4	150	30	130	10	<10	10	4	100	Quartz vein
17	G5	360	20	100	75	<10	10	10	150	Quartz vein
18	G33	45	20	85	180	<10	15	10	375	Milky quartz vein
19	1	250	10	30	10	40			<100	Quartz vein
20	4	800	10	30	20	60			<250	Quartz vein
21	5	1.52%	30	20	20	20			<100	Quartz vein
22	6	500	20	30	10	30			<100	Quartz vein
23	13	2.32%	50	20	10	30			120	Quartz vein
24	18	900	<10	30	10	30			<100	Quartz vein

All values
in ppm, Au
in ppb

ANNEXURE-III									
CHEMICAL ANALYSIS OF STREAM SEDIMENTS SAMPLES, GOVINDPUR AND BOTHLI AREA, CHANDRAPUR DISTRICT									
Sr. No.	Ref.No.	Cu	Pb	Zn	Co	Ni	Ag	As	Au
1	GS-1	45	73	54	131	90	<1	<5	<100
2	GS-2	42	36	57	36	48	<1	<5	<100
3	GS-3	55	47	36	65	69	<1	<5	<100
4	GS-4	70	75	55	129	125	<1	<5	<100
5	GS-5	36	23	50	17	40	<1	<5	<100
6	GS-6	15	14	54	7	17	<1	<5	<100
7	GS-7	17	17	29	31	21	<1	<5	<100
8	GS-8	22	41	37	9	31	<1	<5	<100
9	GS-9	21	22	44	25	29	<1	<5	<100
10	GS-10	28	13	35	13	32	<1	<5	<100
11	GS-11	56	22	46	30	59	<1	<5	<100
12	GS-12	35	31	46	55	64	<1	<5	<100
13	GS-13	116	35	40	178	146	<1	<5	<100
14	GS-14	25	27	38	38	47	<1	<5	<100
15	GS-16	27	26	47	38	54	<1	<5	<100
16	GS-21	21	10	34	20	39	<1	<5	<100
17	GS-22	21	9	33	17	29	<1	<5	<100
18	GS-23	41	60	50	122	106	<1	<5	<100
19	GS-24	25	37	34	83	65	<1	<5	<100
20	GS-25	30	53	36	150	87	<1	<5	<100
21	GS-26	38	25	40	103	85	<1	<5	<100
22	GS-27	25	32	50	41	49	<1	<5	<100
23	GS-28	34	35	62	66	91	<1	<5	<100
24	GS-29	12	10	17	36	30	<1	<5	<100
25	GS-30	126	25	40	111	108	<1	<5	<100
26	GS-31	79	46	53	134	129	<1	<5	<100
27	GS-32	26	21	46	30	42	<1	<5	<100
28	GS-50	34	42	46	82	79	<1	<5	<100
29	GS-10A	32	11	35	16	37	<1	<5	<100

All values
in ppm,
Au in ppb

ANNEXURE-IV									
CHEMICAL ANALYSIS OF SOIL SAMPLES , BOTHLI AREA, CHANDRAPUR DISTRICT									
Sr. No.	Ref.No.	Cu	Pb	Zn	Co	Ni	Ag	As	Au
1	S3E2	40	40	30	20	30	<1	<1	<100
2	S3E9	60	40	40	20	50	<1	<1	<100
3	S1	0.61%	40	40	30	30	<1	<1	<100
4	S6	60	50	40	20	60	<1	<1	<100
5	N1	150	75	40	175	125	<1	<1	<100
6	S3	60	40	40	20	40	<1	<1	<100
7	E3	60	50	30	20	40	<1	<1	<100
8	E5	50	50	40	30	60	<1	<1	<100
9	E6	40	50	40	20	60	<1	<1	<100
10	E0	0.46%	50	30	40	50	<1	<1	<100
11	S2	375	50	40	50	50	<1	<1	<100
12	N1W2	60	50	20	10	40	<1	<1	<100
13	N1W9	40	50	30	20	50	<1	<1	<100
14	S4E1	60	50	40	20	40	<1	<1	<100
15	S4E7	50	60	50	50	60	<1	<1	<100
16	S5W6	150	60	60	40	75	<1	<1	<100
17	S5W8	30	50	30	20	30	<1	<1	<100
18	S5W4	275	50	30	20	50	<1	<1	<100
19	S6W5	50	60	50	30	50	<1	<1	<100
20	S6W9	20	50	20	20	30	<1	<1	<100
21	S6E2	60	50	40	20	60	<1	<1	<100
22	S6E5	60	60	40	20	50	<1	<1	<100
23	S6E8	30	50	30	10	30	<1	<1	<100
24	S3W4	75	60	50	40	60	<1	<1	<100
25	S3W5	40	60	30	20	30	<1	<1	<100
26	S3W6	40	50	60	20	40	<1	<1	<100
27	S3W7	50	60	40	40	50	<1	<1	<100
28	S3W8	60	60	50	30	50	<1	<1	<100
29	S3W9	30	50	30	10	30	<1	<1	<100
30	S1W1	200	60	30	20	30	<1	<1	<100
31	S1W2	60	60	30	40	40	<1	<1	<100
32	S1W3	60	50	30	20	40	<1	<1	<100
33	S1W4	40	60	40	20	40	<1	<1	<100
34	S1W6	75	75	40	50	75	<1	<1	<100
35	S1W7	30	60	40	10	30	<1	<1	<100
36	S1W8	75	60	30	20	40	<1	<1	<100
37	S1W9	50	60	40	30	75	<1	<1	<100
38	S2E1	100	50	40	20	40	<1	<1	<100
39	S2E2	75	50	40	20	40	<1	<1	<100
40	S2E4	60	60	50	20	40	<1	<1	<100
41	S2E7	50	50	30	20	40	<1	<1	<100
42	S2E8	50	60	40	20	60	<1	<1	<100
43	S2E9	60	60	50	20	60	<1	<1	<100
44	W3	100	60	30	10	30	<1	<1	<100
45	W4	30	50	30	<10	20	<1	<1	<100
46	W5	75	60	40	20	60	<1	<1	<100
47	W7	40	50	30	10	40	<1	<1	<100

All
values in
ppm, Au
in ppb

Sr. No.	Ref.No.	Cu	Pb	Zn	Co	Ni	Ag	As	Au	
48	W9	50	20	75	20	50	<1	<1	<100	
49	S1E1	225	20	40	40	40	<1	<1	<100	
50	S1E2	100	10	50	20	40	<1	<1	<100	
51	S1E3	60	20	40	20	40	<1	<1	<100	
52	S1E4	50	20	40	20	40	<1	<1	<100	
53	S1E5	30	20	30	10	40	<1	<1	<100	
54	S1E6	60	30	50	40	75	<1	<1	<100	
55	S1E7	75	50	50	60	100	<1	<1	<100	
56	S1E8	75	40	50	60	100	<1	<1	<100	
57	S1E9	75	40	50	60	100	<1	<1	<100	
58	N1E3	40	<10	30	10	20	<1	<1	<100	
59	N1E4	100	30	40	50	75	<1	<1	<100	
60	N1E6	50	20	40	20	50	<1	<1	<100	
61	N1E7	50	20	40	20	50	<1	<1	<100	
62	N1E8	40	20	40	20	50	<1	<1	<100	
63	N1E9	40	20	40	20	50	<1	<1	<100	
64	S2W1	125	10	40	20	40	<1	<1	<100	
65	S2W2	60	<10	30	20	40	<1	<1	<100	
66	S2W3	50	10	30	20	30	<1	<1	<100	
67	S2W4	60	20	30	30	40	<1	<1	<100	
68	S2W5	60	10	30	10	50	<1	<1	<100	
69	S2W6	30	<10	30	20	40	<1	<1	<100	
70	S2W8	20	<10	40	20	30	<1	<1	<100	
71	S2W9	30	10	30	50	60	<1	<1	<100	
72	S6E3	60	20	670	30	50	<5	10	130	
73	S6E6	60	20	100	15	25	<5	10	200	
74	S6E7	40	20	110	10	30	<5	<1	200	
75	S5W1	50	10	85	15	25	<5	<1	130	
76	S5W2	60	15	225	25	40	<5	<1	140	
77	S5W3	80	<10	240	40	15	15	<1	120	
78	S5W5	85	<10	100	15	30	20	2	160	
79	S5W7	30	<10	180	15	30	30	2	170	
80	S5	35	<10	170	15	30	50	1	180	
81	S3E8	45	<10	110	15	45	75	2	180	
82	S4E6	40	<10	235	10	40	140	2	200	
83	S1E5	50	30	50	20	50	<1	<1	<100	All values in ppm, Au in ppb
84	S6W1	50	20	40	10	40	<1	<1	<100	
85	S6W2	50	30	40	20	50	<1	<1	<100	
86	S6W3	100	30	50	50	60	<1	<1	<100	
87	S6W7	100	30	40	40	75	<1	2	<100	
88	S6W8	50	20	30	60	75	<1	2	<100	
89	S6E1	60	20	50	20	60	<1	<1	<100	
90	S6E4	60	20	40	20	60	<1	<1	<100	
91	S6E9	50	20	40	30	50	<1	<1	<100	
92	S4	75	20	30	20	40	<1	<1	<100	
93	S4E2	60	20	40	20	40	<1	<1	<100	
94	S4E3	50	20	40	20	40	<1	<1	<100	
95	S4E4	60	20	40	20	40	<1	<1	<100	
96	S4E5	50	20	40	20	50	<1	<1	<100	
97	S4E9	50	20	40	20	50	<1	<1	<100	
98	S4E8	40	20	30	20	40	<1	<1	<100	

Sr. No.	Ref.No.	Cu	Pb	Zn	Co	Ni	Ag	As	Au	
99	S4W2	200	20	40	30	40	<1	<1	<100	
100	S4W1	75	20	30	20	30	<1	<1	<100	
101	S4W4	60	10	30	20	30	<1	<1	<100	
102	S4W6	100	50	40	60	75	<1	<1	<100	
103	S4W7	50	20	30	30	40	<1	<1	<100	
104	S5E2	50	10	40	20	40	<1	<1	<100	
105	S5E3	75	20	50	60	75	<1	<1	<100	
106	S5E4	50	20	40	20	40	<1	<1	<100	
107	S5E5	50	20	40	20	40	<1	<1	<100	
108	S5E6	50	20	40	30	60	<1	<1	<100	
109	S5E7	50	20	40	20	60	<1	2	<100	
110	S5E8	40	10	20	20	30	<1	<1	<100	
111	S5E9	60	20	40	20	50	<1	<1	<100	
112	S5W9	40	10	30	20	30	<1	<1	<100	
113	S5E1	200	20	30	30	50	<1	<1	<100	
114	W2	100	10	30	20	40	<1	<1	<100	
115	W6	40	10	30	20	40	<1	<1	<100	
116	N2E3	50	20	30	20	40	<1	<1	<100	
117	E2	100	20	40	60	75	<1	<1	<100	
118	E4	60	20	30	30	40	<1	<1	<100	
119	E9	50	40	50	75	75	<1	<1	<100	
120	S3	75	10	40	30	30	<1	<1	<100	
121	S3E3	50	20	30	30	30	<1	<1	<100	
122	S3E4	50	20	40	20	40	<1	<1	<100	
123	S3E6	60	20	40	20	40	<1	<1	<100	
124	S3E7	50	20	30	20	40	<1	<1	<100	
125	S3W1	275	10	30	40	40	<1	<1	<100	
126	S3W3	200	20	30	30	40	<1	<1	<100	
127	S3W8	30	10	30	20	30	<1	<1	<100	
128	S3W9	60	30	30	30	60	<1	<1	<100	
129	S2E5	40	10	30	20	30	<1	<1	<100	
130	S2E6	40	20	30	20	30	<1	<1	<100	
131	N1W4	40	20	40	10	40	<1	<1	<100	
132	W1	200	<10	35	30	40	10	2	<100	
133	W2	100	10	35	15	30	<5	<1	<100	
134	W6	40	20	30	15	30	<5	<1	<100	
135	W8	40	25	55	10	35	10	<1	<100	
136	N1W1	70	20	60	30	40	<5	<1	<100	
137	N1W3	55	35	50	20	50	<5	<1	<100	
138	N1W4	45	15	40	10	35	<5	<1	<100	
139	N1W5	30	20	25	15	25	<5	<1	<100	
140	N1W6	40	15	35	30	15	<5	10	<100	
141	N1W7	45	25	50	40	55	<5	<1	<100	
142	N1W8	25	15	25	15	15	<5	<1	<100	
143	E1	180	20	40	30	50	<5	<1	<100	
144	E2	100	20	70	50	55	<5	<1	<100	
145	E4	50	10	40	15	30	<5	<1	<100	
146	E7	40	10	40	20	35	<5	<1	<100	
147	E8	85	35	40	115	80	<5	<1	<100	
148	E9	45	35	60	80	60	<5	2	<100	
149	N1E1	100	20	25	30	40	<5	2	<100	

All
values in
ppm, Au
in ppb

Sr. No.	Ref.No.	Cu	Pb	Zn	Co	Ni	Ag	As	Au	
150	N1E2	75	20	45	20	40	<5	<1	<100	
151	N1E5	95	20	40	30	35	<5	1	<100	

ANNEXURE-V										
CHEMICAL ANALYSIS OF BED ROCK SAMPLES, BOTHLI AREA, CHANDRAPUR DISTRICT										
Sr. No.	Ref.No.	Cu	Pb	Zn	Co	Ni	Ag	As	Au	Lithlogy
1	G2	750	20	75	50	10	<1	4	<100	Quartz mica schist
2	G3	0.13%	20	75	60	10	<1	<1	<100	Quartz mica schist
3	G4	10	30	175	60	10	<1	2	<100	Quartz mica schist
4	G5	0.25%	20	75	50	10	<1	1	<100	Quartz mica schist
5	G6	0.11%	20	75	40	10	<1	<1	<100	Silicified sheared Quartzite
6	G7	0.15%	30	100	40	20	<1	<1	<100	Silicified sheared Quartzite
7	G8	0.14%	30	75	40	20	<1	2	<100	Silicified sheared Quartzite
8	G9	0.18%	20	50	40	10	<1	<1	160	Silicified sheared Quartzite
9	G10	10	30	125	60	20	<1	10	140	Silicified sheared Quartzite
10	G11	75	20	150	75	<10	<1	<1	140	Massive felspathic quartzite
11	G12	50	20	125	100	<10	<1	<1	120	Quartz reef
12	G13	30	30	75	50	10	<1	2	120	Quartz reef
13	G18	<10	20	20	275	<10	<1	2	<100	Quartz reef
14	G19	50	50	60	100	30	<1	<1	<100	Quartz reef
15	G20	<10	20	20	125	<10	<1	4	<100	Quartz reef
16	G21	<10	20	20	125	<10	<1	<1	<100	Quartz reef
17	G22	<10	20	20	125	<10	<1	4	<100	Quartz reef
18	G23	<10	20	10	175	<10	<1	<1	<100	Quartz reef
19	G24	<10	20	10	200	<10	<1	10	<100	Quartz reef
20	G-25	<10	40	10	100	10	<1		<100	White ,buffGranite gneiss
21	G-26	<10	20	10	100	10	<1		<100	Highly weathered felspathic gneiss
22	G-27	60	<10	10	225	10	<1		<100	Quartz vein
23	G-28	275	<10	<10	225	<10	<1		<100	Quartz vein
24	G-29	<10	<10	<10	200	<10	<1		<100	Quartz vein
25	G-30	20	10	10	100	<10	<1		<100	Quartz vein
26	G-31	<10	<10	10	100	<10	<1		<100	White pinkish quartzite
27	G-32	<10	10	10	75	<10	<1		150	Quartzite
28	G-33	<10	<10	10	250	<10	<1		100	Felspathic quartzite with limonitic stains
29	G-34	<10	<10	10	100	<10	<1		100	Quartz vein with malachite stains
30	G-35	<10	<10	10	75	<10	<1		100	Quartz vein with malachite stains
31	G-36	20	<10	10	75	<10	<1		100	Quartz vein with malachite stains
32	1	50	50	10			<1		<100	Quartz reef
33	2	75	60	<10			<1		<100	Quartz reef
34	S3	10	50	20			<1		<100	Quartz reef

All values
in ppm,
Au in ppb

ANNEXURE-VI**CHEMICAL ANALYSIS OF PIT/TRENCH SAMPLES , GOVINDPUR AREA,
CHANDRAPUR DISTRICT**

Sr. No.	Ref.No.	Cu	Pb	Zn	Co	Ni	Ag	As	Au	Lithology	All values in ppm and Au values in ppb
1	T1-1	809	11	74	39	53	<1	<5	100	Quartz vein	
2	T1-3	599	7	32	25	53	<1	<5	140	Quartz vein	
3	T2-1	731	16	49	42	46	<1	<5	<100	Quartz vein	
4	T2-3	895	41	65	75	123	<1	<5	140	Quartz vein	
5	T3-1	52	14	84	20	30	<1	<5	<100	Quartz vein	
6	T3-2	55	12	109	21	59	<1	<5	120	Quartz vein	
7	T3-5	100	18	169	35	100	<1	<5	<100	Quartz vein	

**CHEMICAL ANALYSIS OF PIT/TRENCH SAMPLES , BOTHLI AREA,
CHANDRAPUR DISTRICT**

Sr. No.	Ref.No.	Cu	Pb	Zn	Co	Ni	Ag	As	Au	Lithology	All values in ppm and Au values in ppb
1	P1	225	20	100	40	10	<1	4	<100	Milky quartz vein	
2	P2	10	20	100	75	10	<1	2	<100	Milky quartz vein	
3	P3	20	20	100	75	<10	<1	10	<100	Milky quartz vein	
4	P4	<10	10	125	50	<10	<1	2	<100	Millky quartz vein	

FIELD PHOTOGRAPHS OF GOVINDPUR AND BOTHLI COPPER PROSPECT, CHANDRAPUR DISTRICT, MAHARASHTRA**(F.S. 2008-2009)**

Fig-1.
Photograph showing well section of highly weathered granite gneiss near Shirpur area, Chandrapur District (55P/11).



Fig-2.
Photograph showing quartz veins, Shirpur village, Chandrapur District (55P/11).



Fig-3.
Photograph showing quartz vein quarry comprising chalcopyrite, pyrite in silicified zone, near Bothli village, Chandrapur District (55P/11).



Fig-4.
Photograph showing dumped material of quartz vein quarry near Bothli village, Chandrapur District (55P/11).



Fig-5.
Photograph showing Malachite impregnation within metasedimentary rock , near Bothli old working area, Chandrapur District (55P/11).



Fig-6.
Photograph showing old working excavated for copper near Bothli area, Chandrapur District (55P/11).



Fig-7.
Photograph showing crushed material with Malachite impregnation near Bothli old working area, Chandrapur District (55P/11).



Fig-8.
Trenching was done by DGM for copper prospecting in Bothli old working area, Chandrapur District (55P/11).



Fig-9.
Photograph show Sulphides mineralisation in the silicified zones at Bothli Chandrapur District (55P/11).



Fig-10.
Photograph showing a pits excavated by DGM, Nagpur, for copper prospecting near Bothli area, Chandrapur District (55P/11).



Fig-11.
Photograph showing a vegetative cover of Sedimentary rock equivalent to Penganga Group near Bothli area, Chandrapur District (55P/11).



Fig-12.
Photograph showing foliation and lineation in silicified zone within Bengpal Gneiss in Govindpur old working area, Chandrapur District (55P/11).

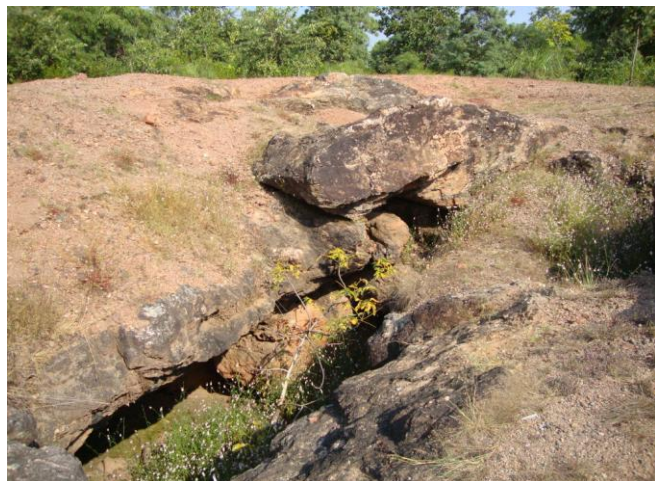


Fig-13.
Photograph showing eastern part of Govindpur old working area, Chandrapur District (55P/11).



Fig-14.
Photograph showing shaft in northern part of Govindpur old working area, Chandrapur District (55P/11).



Fig-15.
Photograph showing dumped material covering main old working of Govindpur , Chandrapur District (55P/11).



Fig-16.
Photograph showing a close view of main old working area near Govindpur village , Chandrapur District (55P/11).



Fig-17.
Photograph showing malachite encrustation in silicified zones confined to enclaves of metapelites within Bengpal gneisses.



Fig-18.
Photograph showing shaft in Western part of Govindpur old working area, Chandrapur District(55P/11).



Fig-19.
Photograph showing the mineralised quartz vein comprising of chalcopyrite, pyrite trending N-S and broadly folded with low plunge towards NE.

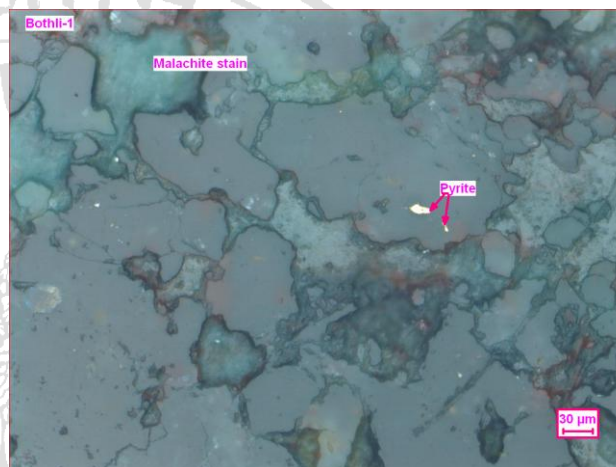


Fig-20.
Micro-photographs of copper ore, showing malachite stains with pyrite grains in Bothli area, Chandrapur district,

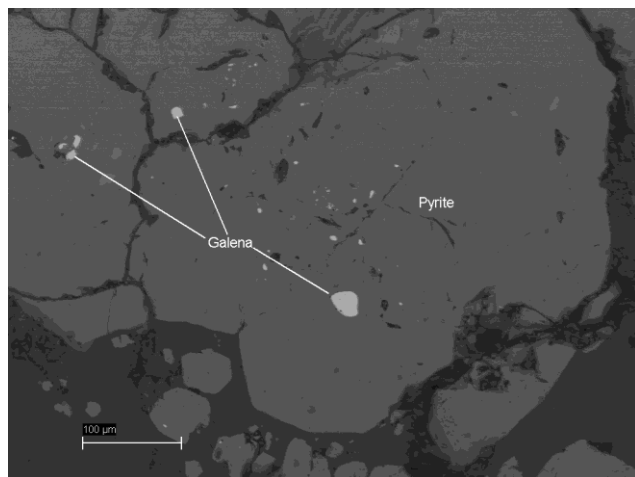


Fig-21.
Micro-photographs of copper ore, showing Galena grains within pyrite. Location: Old working, Govindpur areas, Chandrapur district, Maharashtra.

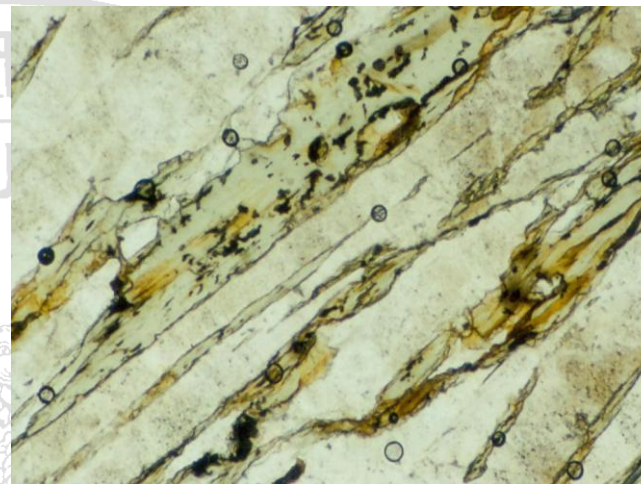


Fig-22.
Development of quartz Ribbons with chlorite in Quartz-chlorite mylonite (10x obj. ppl)

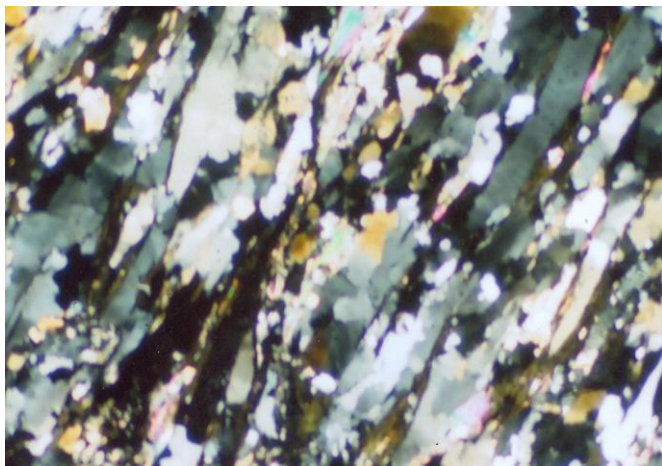


Fig-23.
Development of granulation and recrystallisation in
Quartz-chlorite mylonite (1x obj. under X nicol)

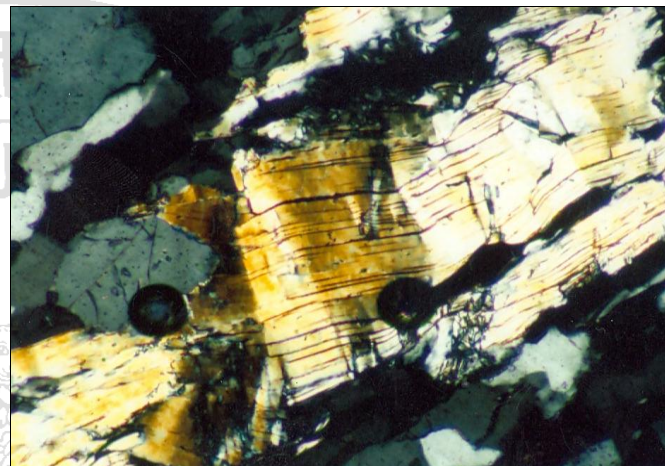


Fig-24.
Kinked muscovite in Quartz-chlorite mylonite (10x obj. under
X nicol)

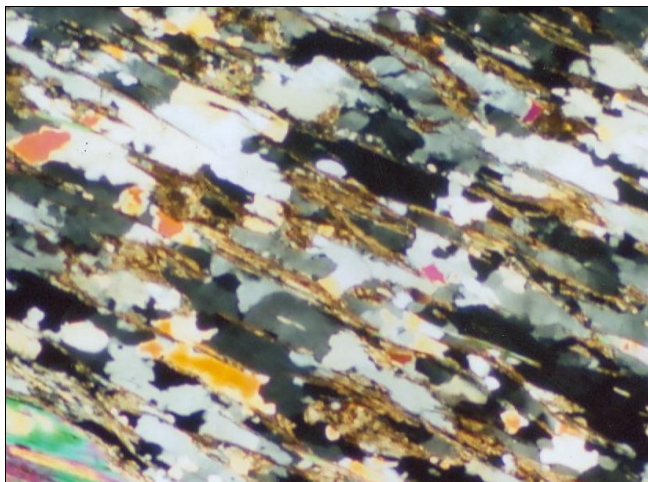


Fig-25.
Development of lenses/ribbons of quartz with
fibrolite in Quartz-fibrolite mylonite (1x obj. Under
Xnicol)

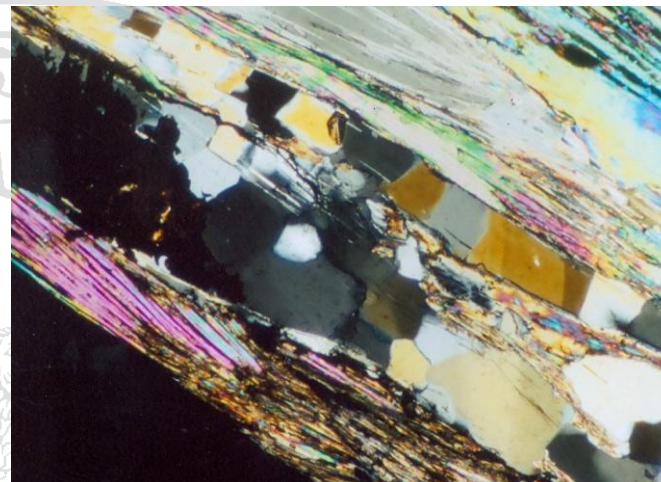


Fig-26.
Fibrolite is replaced by muscovite in Quartz-fibrolite mylonite
(4x obj. under Xnicol)

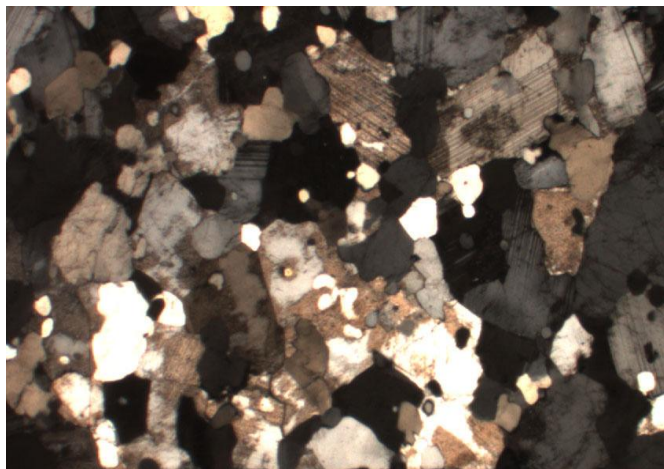


Fig-27.
Hypidiomorphic texture in Garnetiferous quartz monzonite (2.5x under Xnicol)

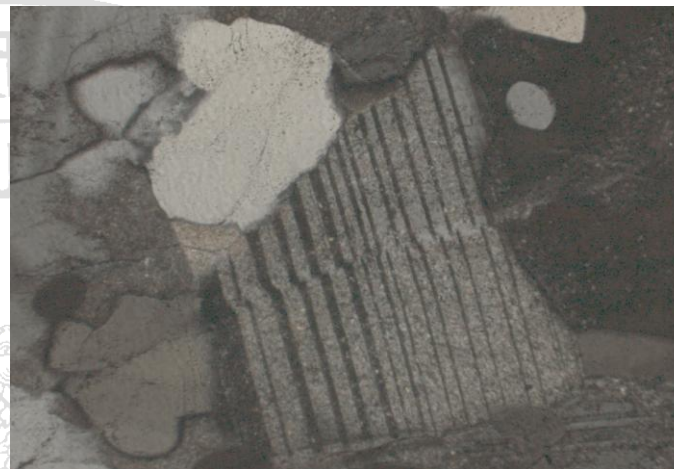


Fig-28.
Distorted and displaced twin lamellae along the micro fracture in Garnetiferous quartz monzonite (10x under Xnicol)

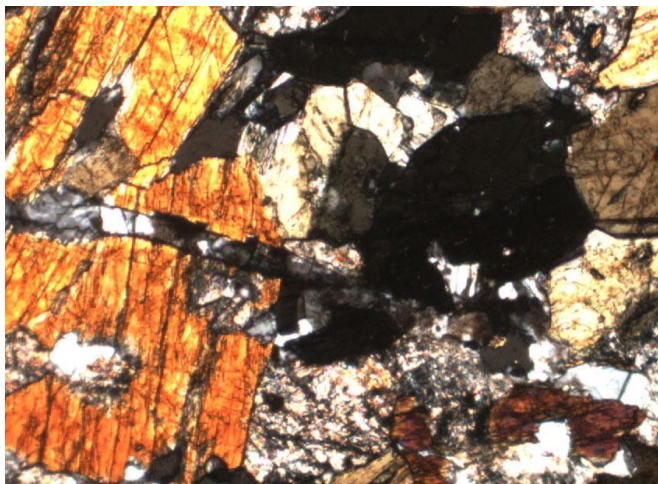


Fig-29.
Quartz is released from plagioclase during sausalitisation
(5x under Xnicol)

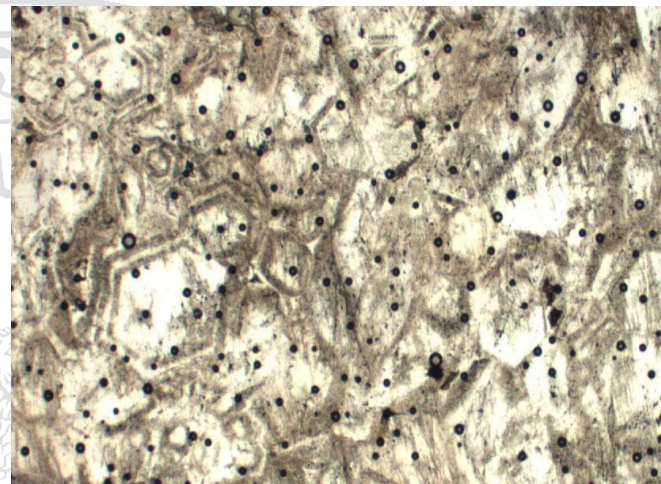


Fig-30.
Hexagonal quartz in quartz vein, (1.6x ppl)

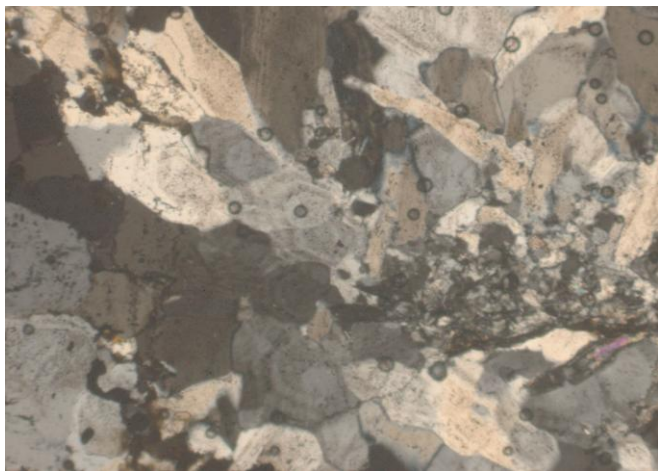


Fig-31.
Quartz crushed into fine aggregates in Quartz vein, (5x
underX nicol)

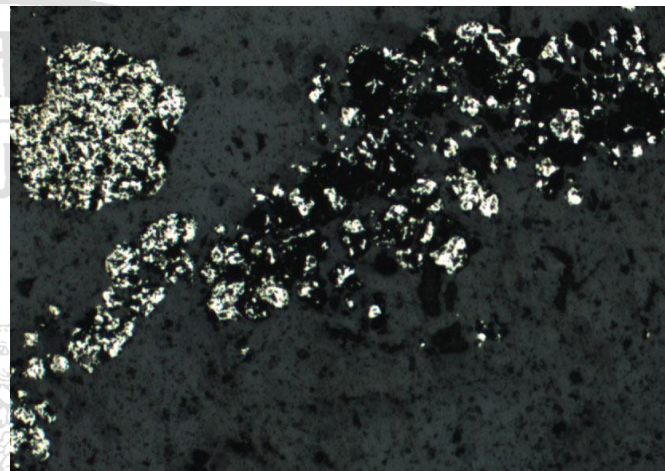


Fig-32.
Fine vein of pyrite in quartz vein, (5x obj. ppl)

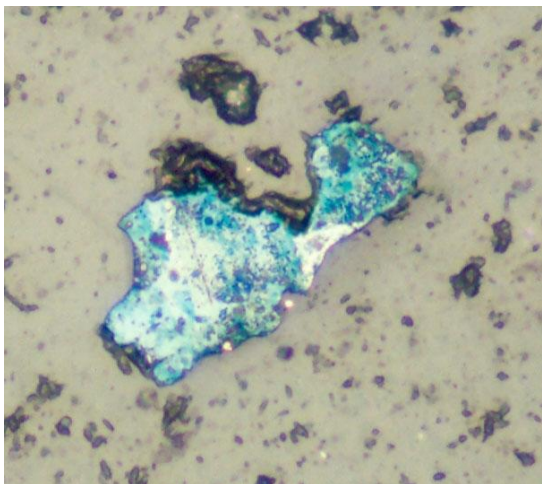


Fig-33.
Covellite in Quartz vein,
(20x obj ppl)

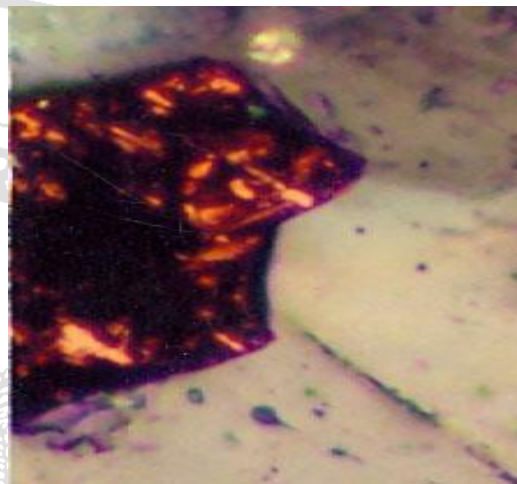


Fig-34.
Covellite showing orange brown anisotropism in Quartz vein
(20x obj under X nicol)

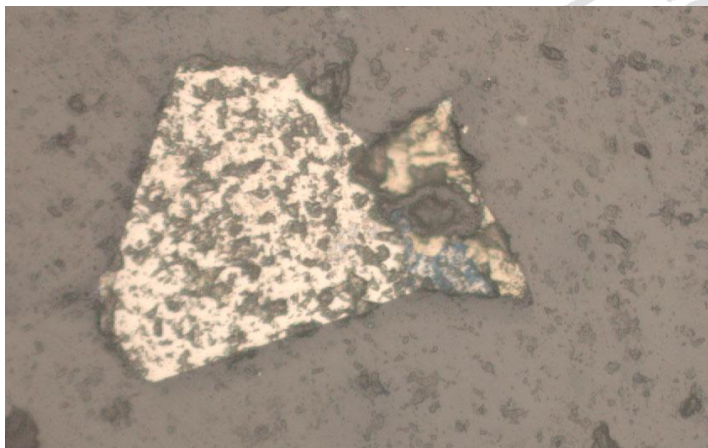


Fig-35.
Chalcopyrite, pyrite, covellite in quartz vein,(20x ppl)

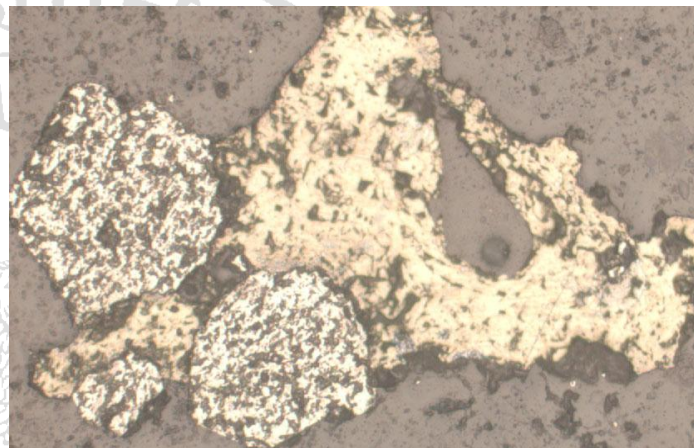
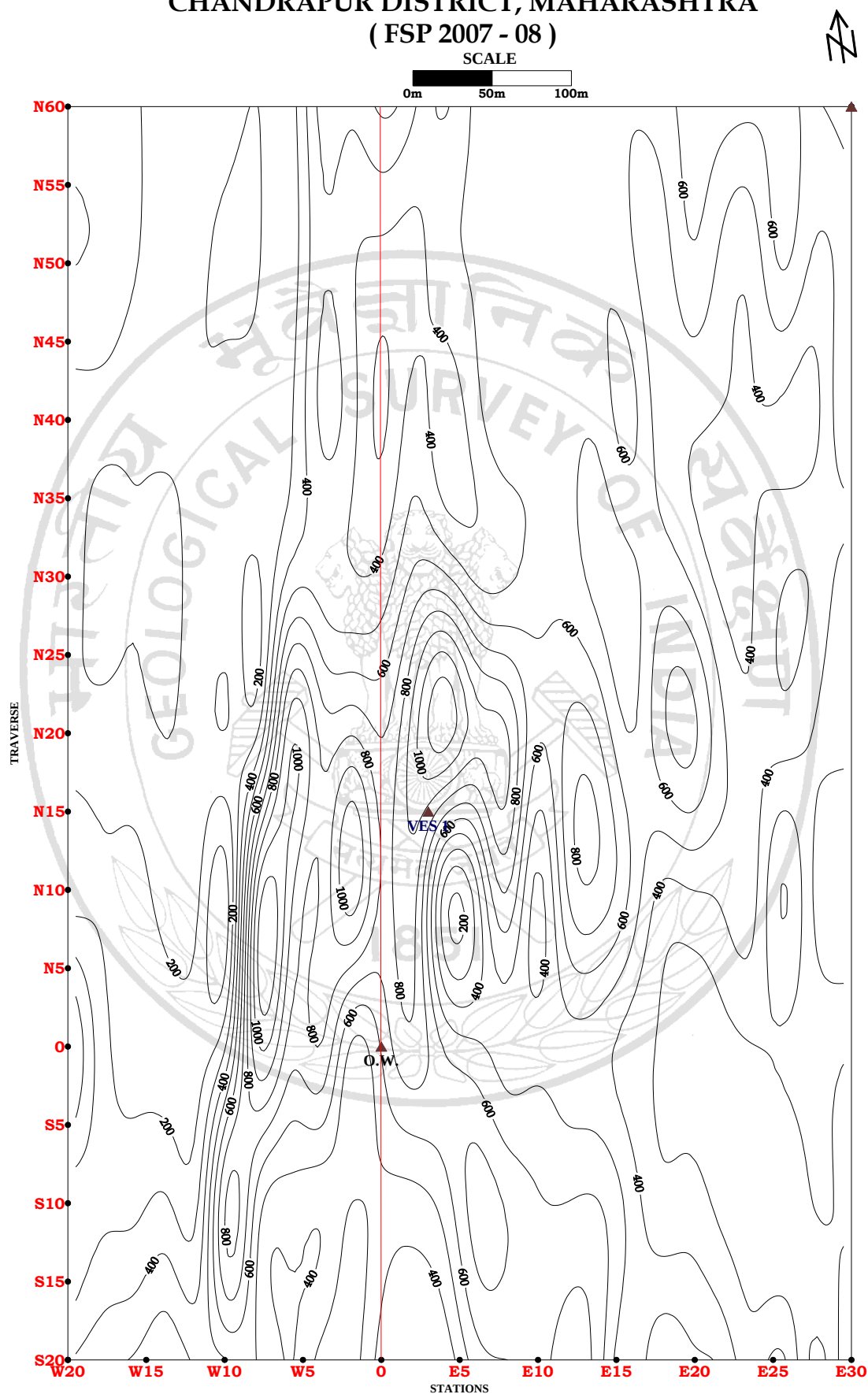
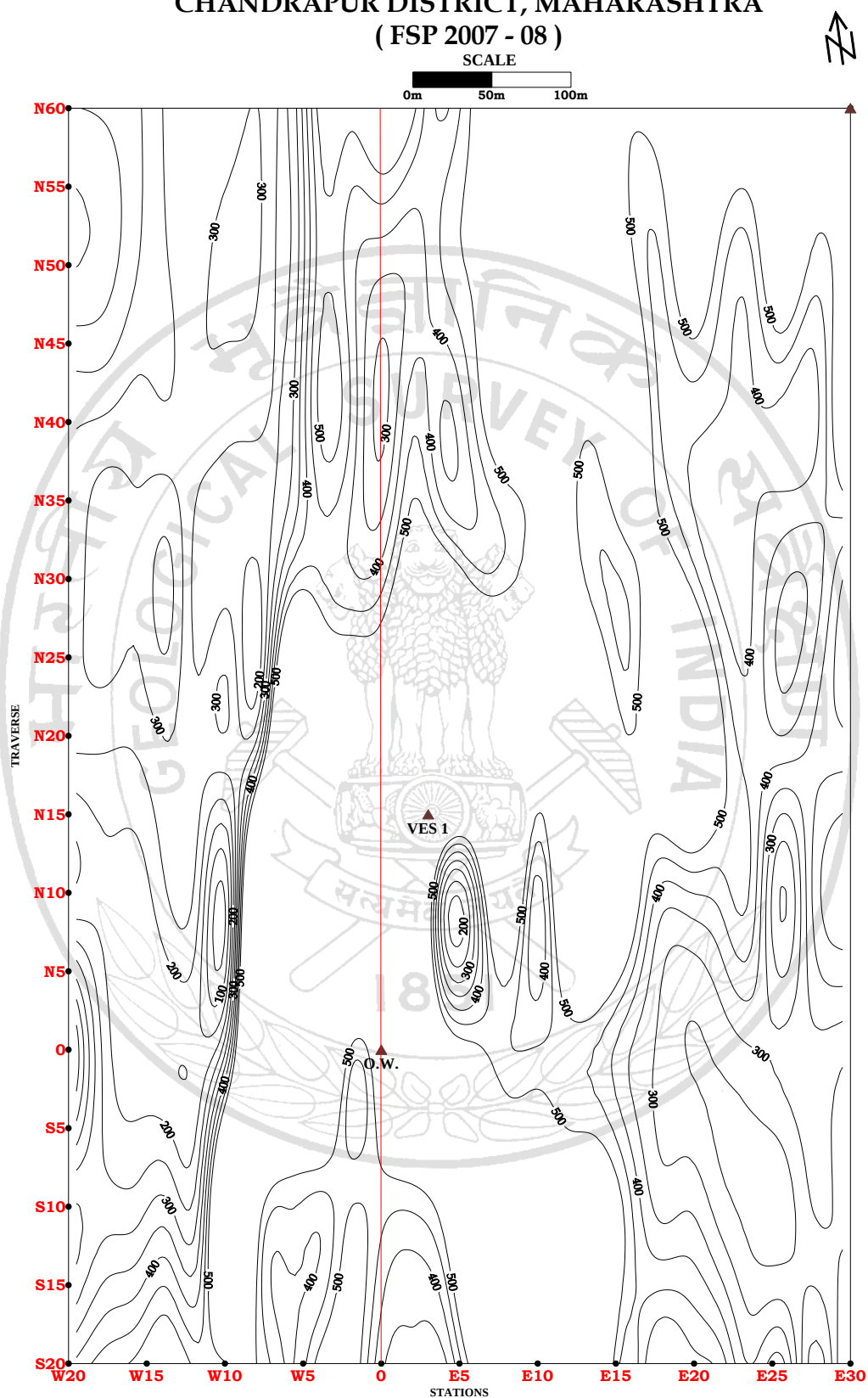


Fig-36.
Pyrite replacing chalcopyrite quartz vein shows replacement texture, Bothli area (10x ppl)

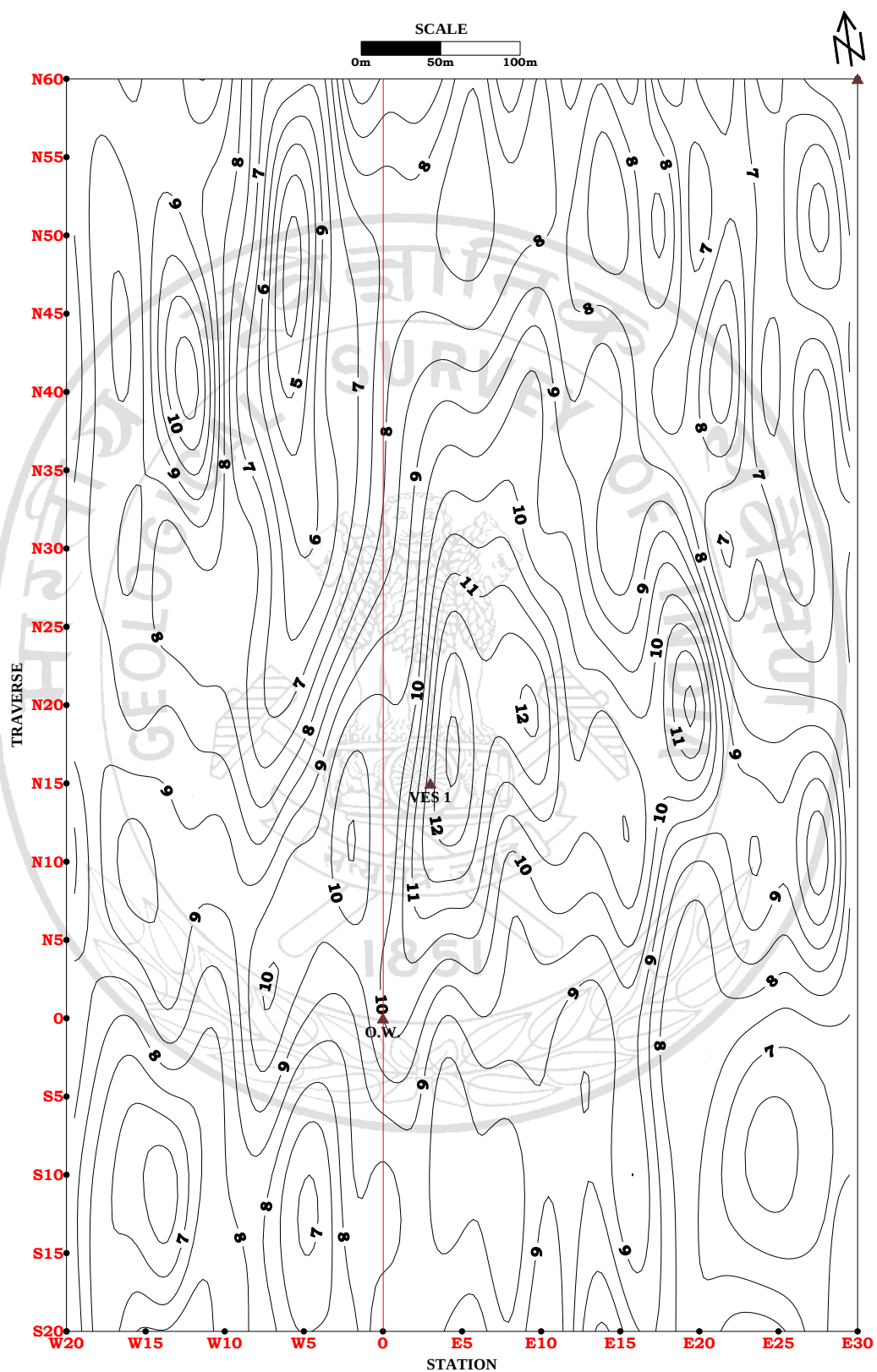
RESISTIVITY CONTOUR MAP OF GOVINDPUR COPPER PROSPECT CHANDRAPUR DISTRICT, MAHARASHTRA (FSP 2007 - 08)



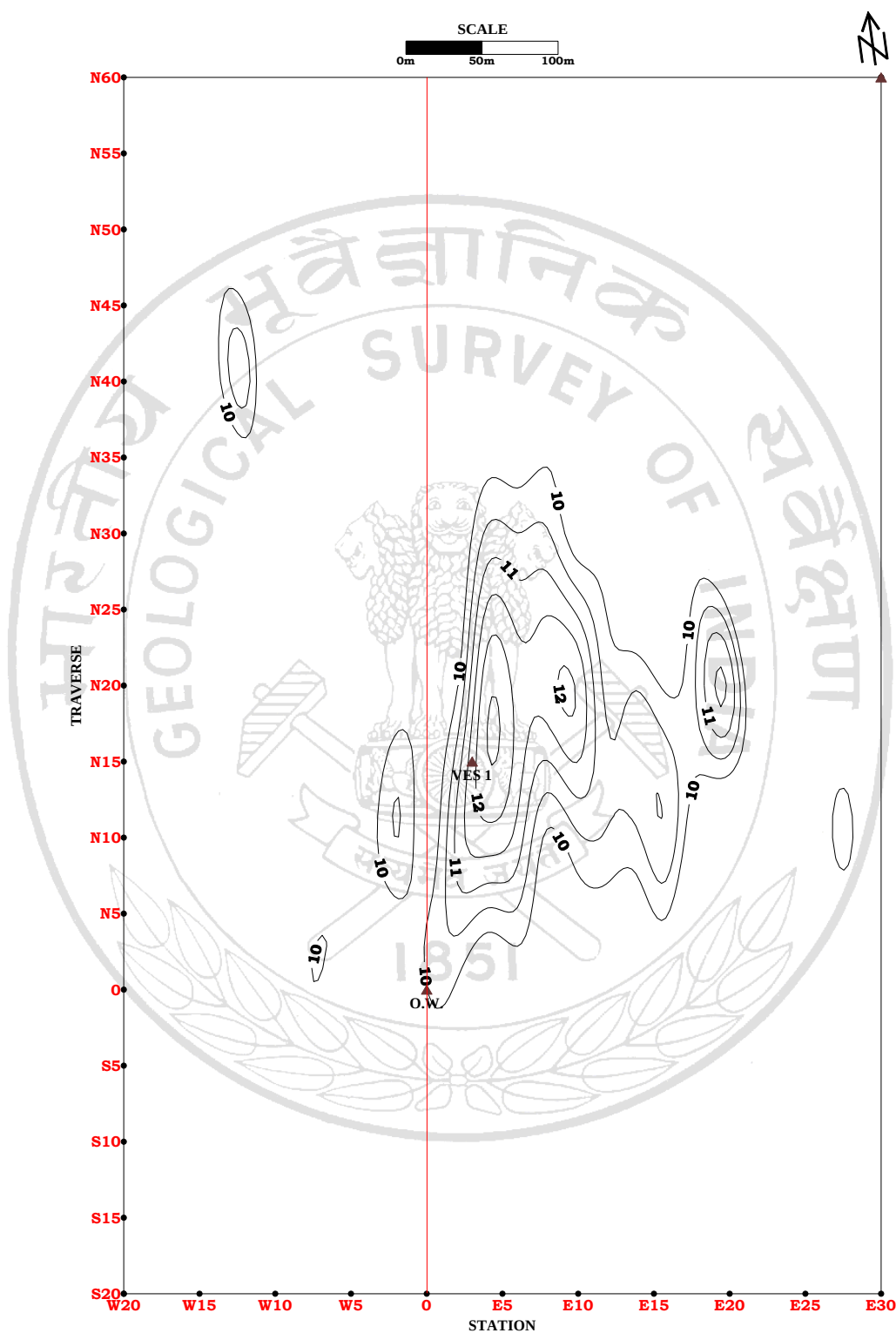
**LOW RESISTIVITY CONTOUR MAP OF GOVINDPUR COPPER PROSPECT
CHANDRAPUR DISTRICT, MAHARASHTRA
(FSP 2007 - 08)**



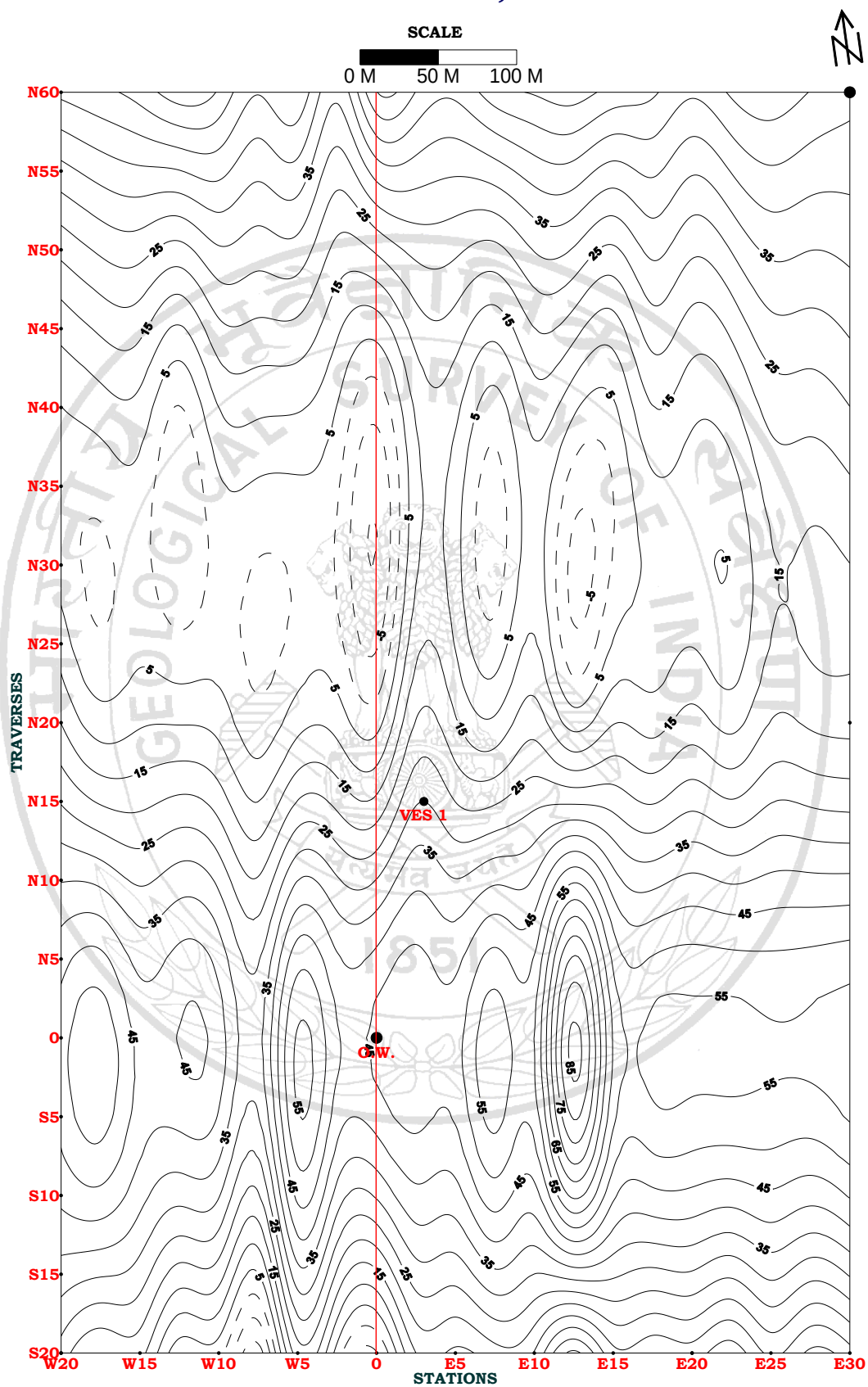
**CHARGEABILITY CONTOUR MAP OF GOVINDPUR COPPER PROSPECT
CHANDRAPUR DISTRICT, MAHARASHTRA
(FSP 2007 - 08)**



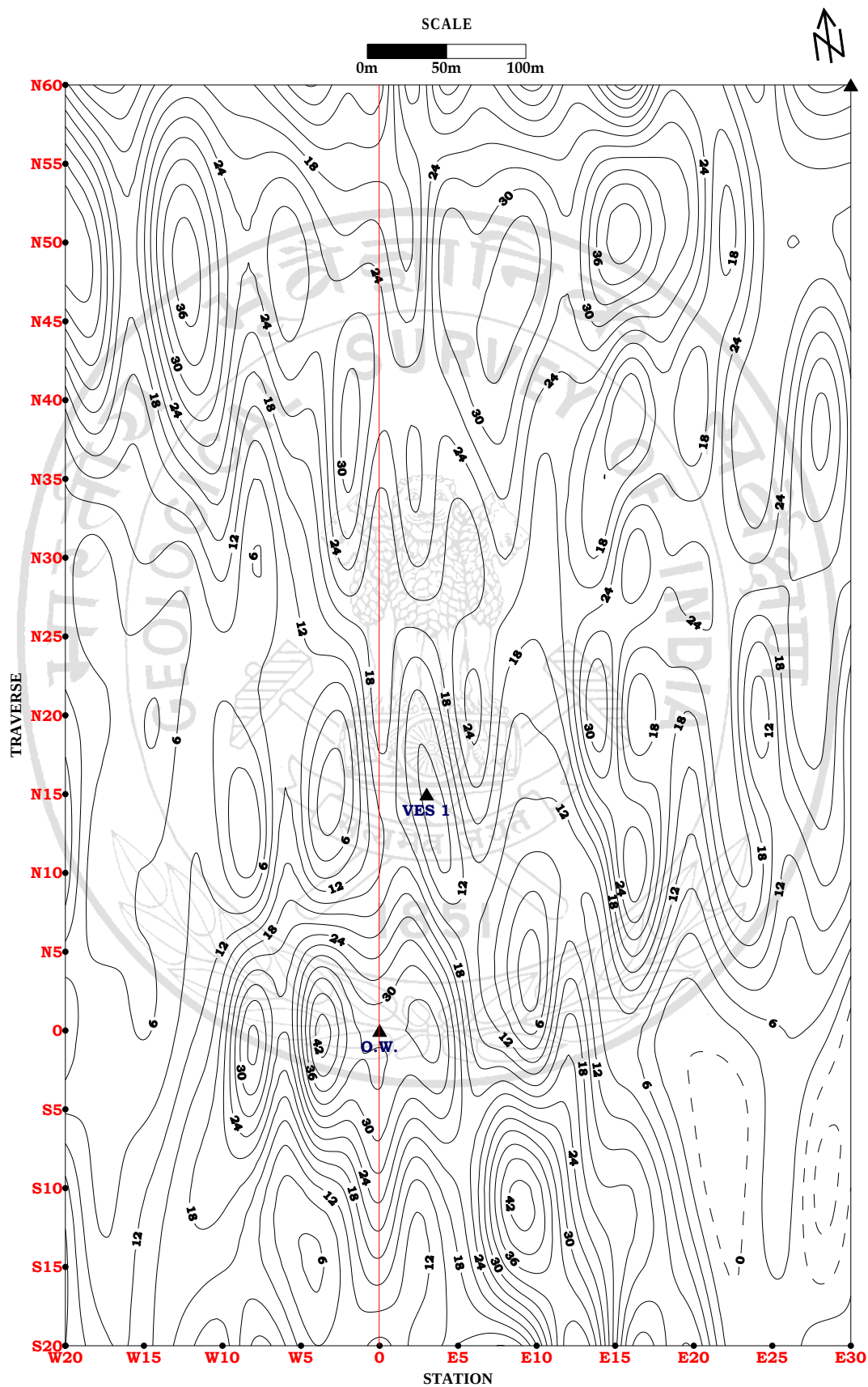
**HIGH CHARGEABILITY CONTOUR MAP OF GOVINDPUR COPPER PROSPECT
CHANDRAPUR DISTRICT, MAHARASHTRA
(FSP 2007 - 08)**



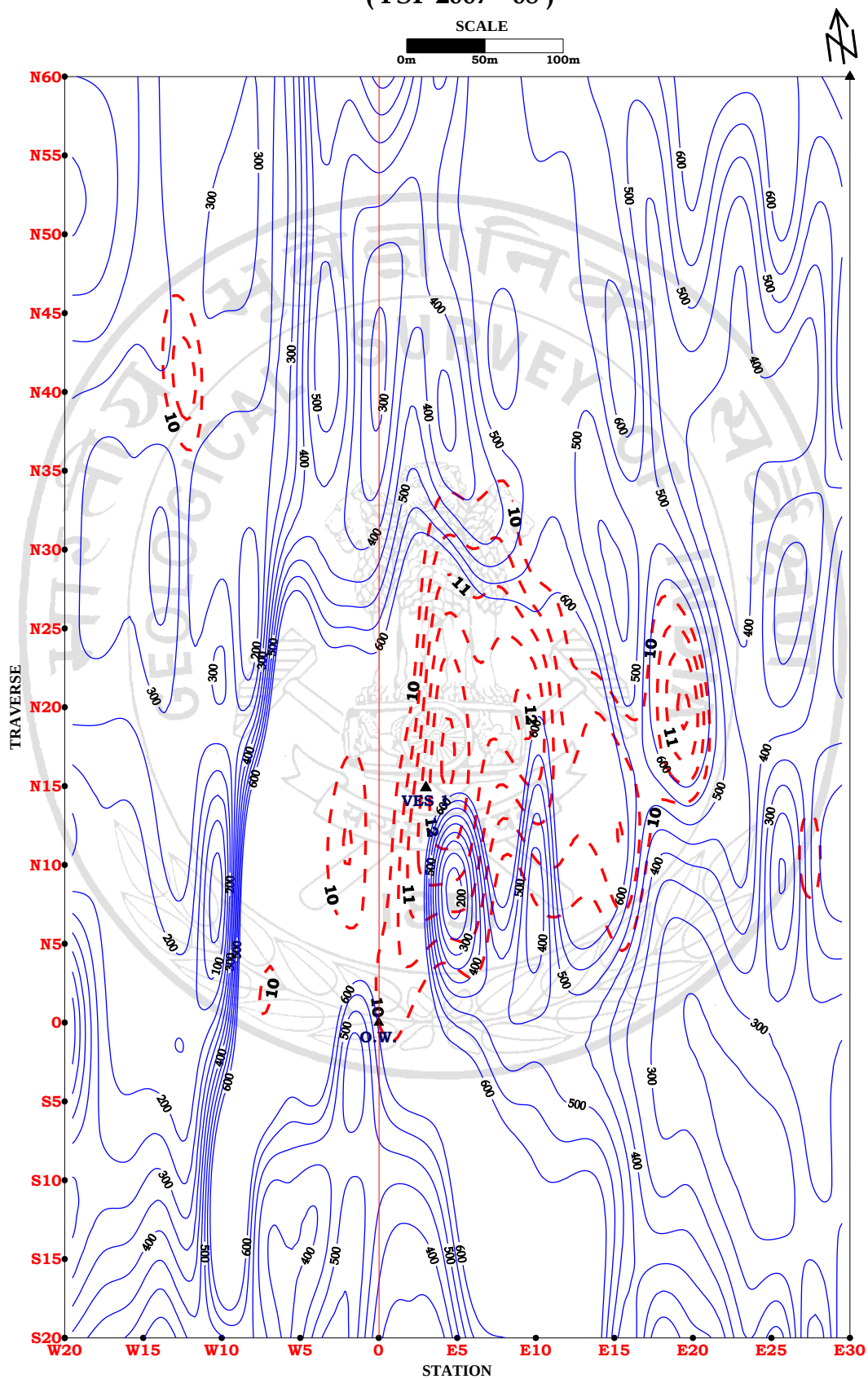
MAGNETIC CONTOUR MAP GOVINDPUR COPPER PROJECT CHANDRAPUR DISTRICT, MAHARASHTRA



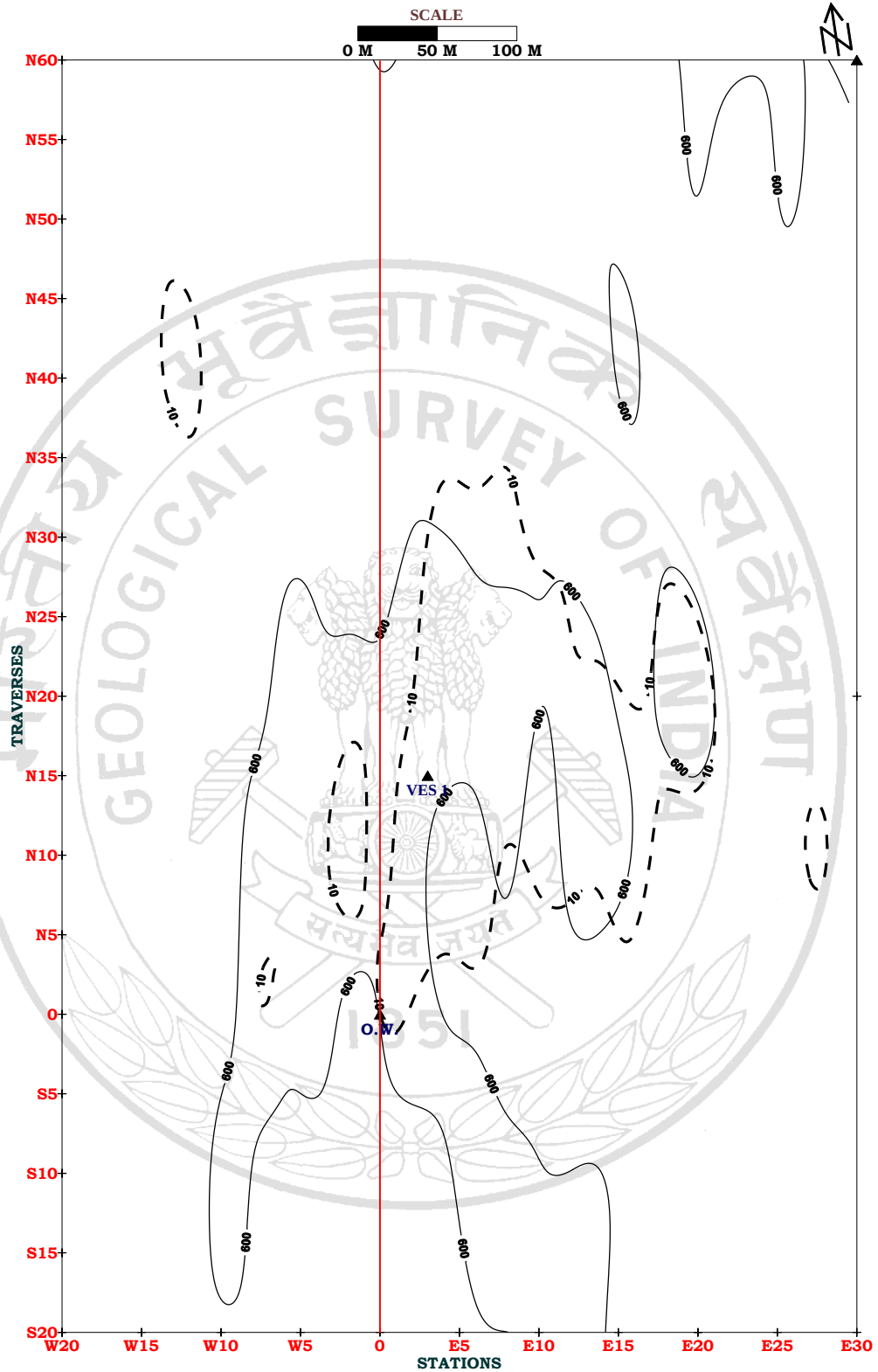
**SP CONTOUR MAP OF GOVINDPUR COPPER PROSPECT
CHANDRAPUR DISTRICT, MAHARASHTRA
(FSP 2007 - 08)**



**HIGH CHARGEABILITY AND LOW RESISTIVITY CONTOUR
GOVINDPUR COPPER PROSPECT
CHANDRAPUR DISTRICT, MAHARASHTRA
(FSP 2007 - 08)**



**COMBINED MAP OF HIGH RESISTIVITY AND HIGH CHARGEABILITY
GOVINDPUR COPPER PROJECT, CHANDRAPUR DISTRICT, MAHARASHTRA**



**COMBINED MAP OF LOW S.P. AND HIGH CHARGEABILITY
GOVINDPUR COPPER PROJECT, CHANDRAPUR DISTRICT, MAHARASHTRA**

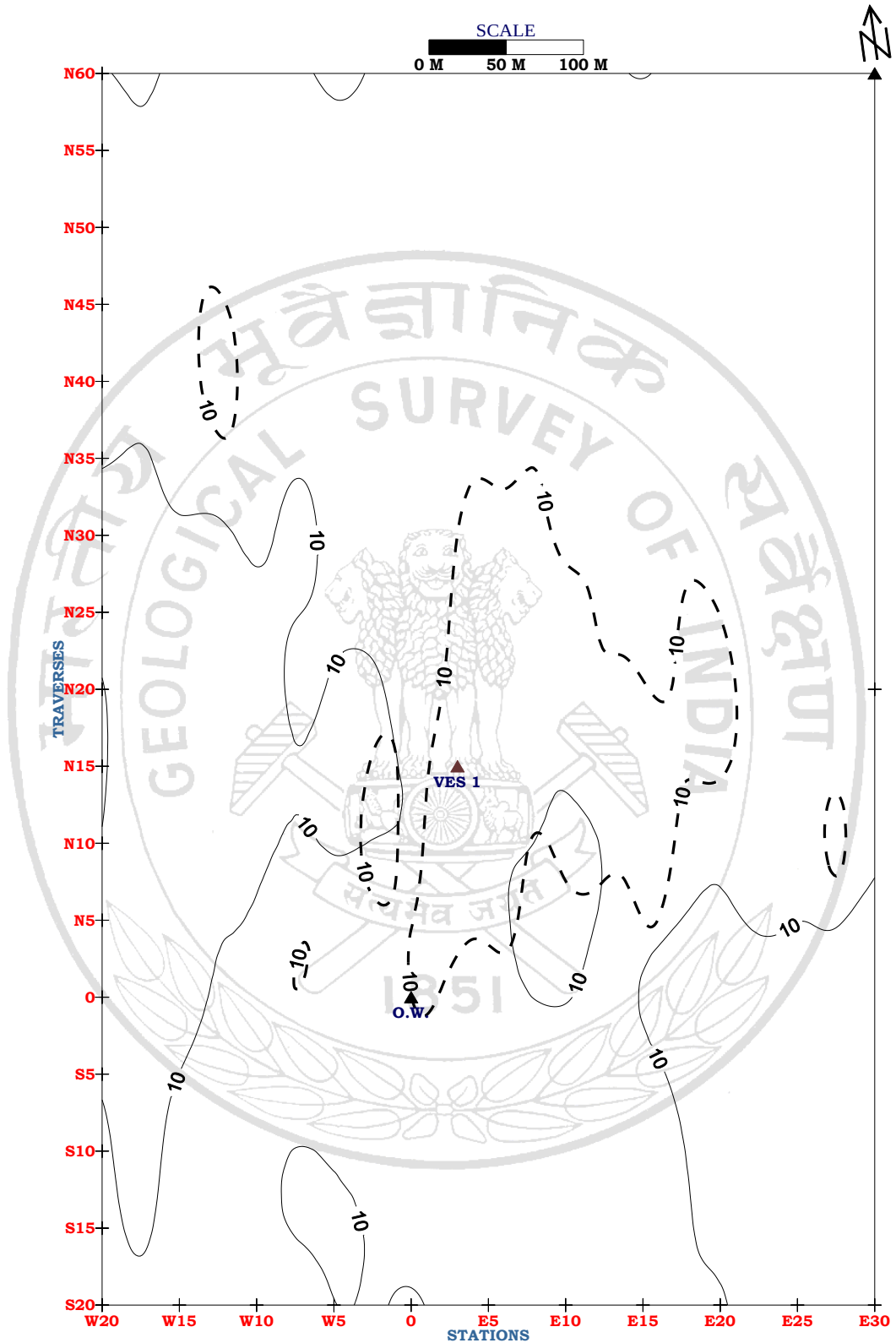


Fig.46

